Turbocharging Hybrid Scheme for Rough Volatility Model Simulation

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Abstract

We explore the fast and accurate Monte Carlo simulation of the stochastic Volterra equations, with the goal to accelerate option pricing and implied volatility calculation in rough volatility models. Leveraging the Markovian approximation process briefed in Abi Jaber (2018) as the precomputation step, we present the Accelerated Hybrid Scheme, a novel simulation algorithm that uses the fast Fourier transform to accelerate the classical hybrid scheme of Bennedsen et al. (2017). Under globally Lipschitz coefficients with linear growth, this algorithm achieves a strong convergence result and a time complexity of $\mathcal{O}(n\cdot(n_f+\log n))$ for n time steps and n_f factors that approximates the Laplace transform of the kernel. We validate our algorithm through numerical experiments on the rough Heston model, demonstrating its effectiveness even with non-Lipschitz coefficients. In contrast with existing simulation methods, the Accelerated Hybrid Scheme provides a fast solution for rough volatility models with more numerical stability.

Keywords: rough volatility models, stochastic Volterra equations, Monte Carlo simulation, option pricing, implied volatility.

1 Introduction

Stochastic volatility models have dominated the research and practice in financial methematics for decades, providing a general framework for option pricing, hedging, and volatility forecasting. The majority of these models feature a price process S_t in a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ of the form

$$dS_t = \mu S_t\, dt + \sigma_t S_t\, dW_t,$$

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where μ is a drift term, σ_t is the volatility process, and W_t is a standard Brownian motion. The volatility process σ_t is typically modeled as a Brownian semimartingle, such as the Heston model Heston (1993), the SABR model Hagan et al. (2002), and the 3/2 model. These models have been widely used due to their tractability and ability to capture the volatility skew, but their volatility trajectories with regularity close to that of Brownian motion deviate from some empirical properties such as long memory properties or roughness, as described in Gatheral et al. (2018).

Due to the smoothness of volatility being questioned by empirical evidence, rough volatility models have emerged to better characterize the behavior observed in realized volatility time series. However, these models deviate from classical assumptions such as the Brownian semimartingale and Markov properties, which complicates the derivation of closed-form solutions and impedes efficient numerical simulation.

Since the seminal work on rough volatility models, the literature has broadly classified these models into two primary categories: those driven by fractional Brownian motion (fBM) and those given by stochastic Volterra equations. In this paper, we focus on the simulation of the latter class, which is generally of the form

$$X_t = X_0 + \int_0^t K(t-s)b(X_s)\,ds + \int_0^t K(t-s)\sigma(X_s)\,dW_s, \tag{1}$$

where W is a multi-dimensional Brownian motion, and the kernel K and coefficients b and σ are functions of certain regularity and integrability. The solutions of such models are typically adapted processes defined on some probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions, see Abi Jaber (2018). In particular, we consider the simulation of one-dimensional stochastic Volterra equations, which are a special case of (1) with b and σ being scalar functions, and W being a one-dimensional standard Brownian motion. Then, we extend our algorithm to the rough Heston model, whose coefficients are given by $\sigma(x) = \nu\sqrt{x}$ and $b(x) = \lambda(\theta - x)$ in (1).

Contributions. While various simulation algorithms have been proposed for stochastic Volterra equations, the existing methods are either computationally expensive or lack theoretical convergence guarantees. In this paper, we propose an efficient and accurate simulation algorithm for the stochastic Volterra equations based on the hybrid scheme introduced in Bennedsen et al. (2017). The idea of our algorithm is to precompute the Markovian approximation process and use the fast Fourier transform to accelerate the hybrid scheme simulation, which achieves a great balance between accuracy and efficiency. We establish the convergence of our algorithm under the global Lipschitz assumptions on coefficients, and validate it through numerical experiments on the rough Heston model that demonstrate its effectiveness with Lipschitz conditions relaxed.

In general, our simulation algorithm exhibits three main advantages. First, it is computationally efficient with a time complexity of $\mathcal{O}(n\cdot(n_f + \log n))$ for n time steps and n_f factors that approximate

the Laplace transform of the kernel. Second, it achieves a strong convergence result under globally Lipschitz coefficients with linear growth. Such theoretical properties are hard to achieve under the existing hybrid scheme, where the update of the stochastic Volterra equation is dependent on all of its history. Third, it is broadly applicable to most rough volatility models, including the generalized Heston-type models introduced by Gao (2024):

$$dS_t = (r-d)S_t\,dt + \sqrt{V_t}S_t\,dW_{1t} \tag{2}$$

$$\begin{aligned} V_t = V_0 + \varphi_0(t) + \int_0^t \lambda(\theta - V_s)\,ds + \bar{\gamma}_1 \int_0^t (t-s)^{H_1-1/2}\sqrt{V_s}\,ds + \bar{\gamma}_2 \int_0^t (t-s)^{H_2-1/2}\sqrt{V_s}\,ds \\ + \gamma_1 \int_0^t (t-s)^{H_1-1/2}\sqrt{V_s}\,dW_{1s} + \gamma_2 \int_0^t (t-s)^{H_2-1/2}\sqrt{V_s}\,dW_{2s}, \end{aligned} \tag{3}$$

where φ_0 is a deterministic function, W_1, W_2 are two independent standard Brownian motions, and $H_1, H_2 \in (0, 1/2)$. These models are widely used in option pricing and volatility forecasting in the financial market.

The rest of this paper is organized as follows. Section 2 and 3 introduce the simulation problem we focus on and review the popular simulation techniques for Volterra integral equations and stochastic volatility models, along with their theoretical foundations. Based on the limitations of existing methods, we propose our accelerated simulation algorithm in Section 4 and present the theoretical properties supporting its implementation. Section 5 provides numerical results demonstrating the efficiency and accuracy of our simulation technique, with comparisons to various references and benchmarks. Finally, Section 6 concludes the paper.

2 Problem setting

In following sections, we focus on the Monte Carlo simulation of the one-dimensional stochastic Volterra equation given by

$$X_t = X_0 + \int_0^t K(t-s)b(X_s)\,ds + \int_0^t K(t-s)\sigma(X_s)\,dW_s \tag{4}$$

defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$, where W is a standard Brownian motion, X_0 is a given constant, $K : \mathbb{R}_+ \to \mathbb{R}_+$ is a possibly singular kernel, and $b : \mathbb{R} \to \mathbb{R}$, $\sigma : \mathbb{R} \to \mathbb{R}$ are globally Lipschitz continuous functions with linear growth.

In what follows, we adopt the assumptions on the kernel K as given in Bennedsen et al. (2017) (see Assumption 1). The conditions imposed on the coefficients b and σ are stated in detail in Assumption 2.

Assumption 1. *The kernel function K in (4) is continuously differentiable on $(0, \infty)$, and satisfies the following properties:*

(i) there exist constants $\alpha \in (-\frac{1}{2}, \frac{1}{2}) \setminus \{0\}$ *and* $L_K : (0,1] \to \mathbb{R}_+$ *such that*

$$K(x) = x^\alpha L_K(x), \quad x \in (0,1],$$

where L_K *is a slowly varying function at 0 and bounded away from 0. Moreover,*

$$\exists C > 0,\ \forall x \in (0,1],\ |L_K'(x)| \le C(1 + x^{-1}).$$

(ii) the derivative K' *is ultimately monotonic with* $\int_1^\infty K'(x)^2 dx < \infty$.

(iii) For some $\beta \in (-\infty, -\frac{1}{2})$, $K(u) = \mathcal{O}(u^\beta)$ *as* $u \to \infty$.

Under Assumption 1, we are particularly interested in the case where the kernel K is a power-law function, i.e., $K(u) = \frac{u^{H-1/2}}{\Gamma(H+1/2)}$ with $H \in (0, \frac{1}{2})$. This choice of kernel leads to rough sample paths and is commonly used in rough volatility models.

Assumption 2. *The coefficients* b *and* σ *in (4) are measurable functions that satisfy*

(i) global Lipschitz continuity, i.e., there exist constant $L > 0$ *such that*

$$|b(x) - b(y)| + |\sigma(x) - \sigma(y)| \le L|x - y|, \quad \forall x, y \in \mathbb{R}.$$

(ii) the linear growth condition, i.e., there exist constant $C > 0$ *such that*

$$|b(x)| + |\sigma(x)| \le C(1 + |x|), \quad \forall x \in \mathbb{R}.$$

Following Assumption 2, along with the integrability condition

$$\int_0^T \left(|K(s)| + |K(s)|^2\right) ds < \infty \tag{5}$$

for any $T > 0$, which is implied by Assumption 1, X_t in (4) admits a unique continuous strong solution up to time T (see Theorem 3.1, Zhang (2010)).

In addition to Assumption 1, if we further assume that the kernel function satisfies Assumption 3, we can establish the weak existence of the solution to the stochastic Volterra equation (4) without the global Lipschitz condition on the coefficients b and σ.

Assumption 3. *There exists* $\gamma > 0$ *and* $C > 0$ *such that for any* $t, h \ge 0$ *with* $t + h \le T$,

$$\int_0^h |K(s)|^2 ds + \int_0^{T-h} |K(h+s) - K(s)|^2 ds \le C h^{2\gamma}.$$

Under the conditions specified in Assumptions 1, 2(ii), and 3, the stochastic Volterra equation (4) admits a weak continuous solution. This result follows from standard arguments in the theory of stochastic Volterra equations (see Abi Jaber (2018) for a detailed discussion).

In the rest of this paper, we will first propose a simulation algorithm for the process (4) and establish its convergence. After that, we will extend the algorithm to the rough Heston model and analyze its performance numerically.

The rough Heston model is given by

$$dS_t = \sqrt{V_t} S_t \, dB_t \tag{6}$$

$$V_t = V_0 + \frac{1}{\Gamma(H+1/2)} \int_0^t (t-s)^{H-1/2} \left(\lambda(\theta - V_s) \, ds + \nu \sqrt{V_s} \, dW_s \right), \tag{7}$$

where the variance process (7) is a special case of the stochastic Volterra equation (4) in which $b(x) = \lambda(\theta - x)$, $\sigma(x) = \nu\sqrt{x}$. B_t and W_t are standard Brownian motions defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$ with a prescribed correlation coefficient $\rho \in [-1, 1]$. The initial variance is given by V_0 and the Hurst index satisfies $H \in (0, \frac{1}{2})$. Moreover, in defining the asset price process (6) under the risk-neutral measure $\mathbb{Q}$, it is assumed, without loss of generality, that the risk-free interest rate r equals the dividend yield d.

Notations. Throughout the paper we adopt the notational conventions as follows. Following Glasserman (2004), we say that a discretization $\widehat{X}$ has a strong order of convergence $\beta > 0$ if

$$\mathbb{E}[|\widehat{X}_{n\Delta t} - X_T|] \leq C(\Delta t)^{\beta}$$

for some constant $C > 0$ and all sufficiently small Δt. β can also be acquired through the rooted mean square error (RMSE) of the discretization via the Lyapunov inequality

$$\mathbb{E}[|\widehat{X}_{n\Delta t} - X_T|] \leq \left(\mathbb{E}[|\widehat{X}_{n\Delta t} - X_T|^2] \right)^{\frac{1}{2}}.$$

We also use the big O notation to denote the order of a function, i.e., $f(x) = \mathcal{O}(g(x))$ as $x \to x_0$ if there exists a constant $C > 0$ such that $|f(x)| \leq C|g(x)|$ for all x in a neighborhood of x_0. When $x \to x_0$ is omitted in the notation, it is understood that the limit is taken as $x \to \infty$.

3 Related simulation methods

There is a rich literature on the simulation of stochastic Volterra equations (1), (4), and (7), which can be roughly classified into two categories: discretization methods and Markovian approximation methods.

3.1 Discretization methods

These methods approximate the stochastic integral by discretizing the time interval $[0, T]$ into small steps Δt. The most straightforward discretization method is the Euler-type scheme as mentioned in

Richard et al. (2021) and Richard et al. (2023). On the discrete-time grid $0 = t_0 < t_1 < \cdots < t_n = T$, the Euler-type scheme simulates the process (4) as

$$\widehat{X}_{t_k} = X_0 + \sum_{j=0}^{k-1} K(t_k - t_j) b(\widehat{X}_{t_j}) \Delta t_{j+1} + \sum_{j=0}^{k-1} K(t_k - t_j) \sigma(\widehat{X}_{t_j}) (W_{t_{j+1}} - W_{t_j}), \tag{8}$$

and the variance process (7) as

$$\widehat{V}_{t_k} = V_0 + \frac{1}{\Gamma(H+1/2)} \sum_{j=0}^{k-1} (t_k - t_j)^{H-1/2} \left(\lambda(\theta - (\widehat{V}_{t_j})_+) \Delta t_{j+1} + \nu \sqrt{(\widehat{V}_{t_j})_+} (W_{t_{j+1}} - W_{t_j}) \right), \tag{9}$$

where $(V_t)_+ := \max(V_t, 0)$, $\Delta t_{j+1} = t_{j+1} - t_j$, $W_{t_{j+1}} - W_{t_j} \sim \mathcal{N}(0, \Delta t_{j+1})$.

The Euler-type scheme is straightforward to implement and demonstrates strong convergence provided that Assumption 2 holds for the coefficients b and σ. The following proposition presents the strong convergence result associated with the Euler scheme.

Proposition 1 (Theorem 2.2, Richard et al. (2021))**.** *Let Assumption 2 holds true for the process* (4)*, and let* $\alpha := \alpha_1 \wedge \alpha_2 > 0$ *such that for all* $0 \le t \le t' \le T$ *and time steps* $n \ge 1$*,*

$$\int_t^{t'} \left(\|K(t' - s)\| + \|K(t' - \eta_n(s))\| \right) ds \le C(t' - t)^{\alpha_1 \wedge 1}$$

and

$$\int_t^{t'} \left(\|K(t' - s)\|^2 + \|K(t' - \eta_n(s))\|^2 \right) ds \le C(t' - t)^{2(\alpha_2 \wedge 1)},$$

where $C > 0$ *is a constant,* $\eta_n(s) := t_k$ *for* $s \in [t_k, t_{k+1})$*,* $k \ge 0$*. Then, there exists a constant* $C_p \in (0, \infty)$ *depending on* T, p, C *such that for all* $t \in [0, T]$ *and* $n \ge 1$*,*

$$\mathbb{E}[|\widehat{X}_t - X_t|^p] \le C_p (1 + \mathbb{E}[|X_0|^p]) \delta_n^{p(\alpha \wedge 1)},$$

where $\delta_n := \max_{0 \le k \le n-1} (t_{k+1} - t_k)$*.*

For kernel $K(u) = \frac{u^{H-1/2}}{\Gamma(H+1/2)}$ and a equal partition of $[0, T]$ with $t_k = k\Delta t$, Proposition 1 implies a strong convergence result of $\mathbb{E}[|\widehat{X}_t - X_t|^p] = \mathcal{O}(n^{-pH})$.

While the Euler scheme admits a strong convergence result under Lipschitz coefficients b, σ, this convergence property is not applicable to the rough Heston model (6)-(7) as $\sigma(x) = \nu\sqrt{x}$ fails to satisfy the global Lipschitz condition. In this case, the Euler scheme only admits a weak convergence result (see Theorem 2.2, Richard et al. (2023)). Moreover, the Euler scheme is limited by the necessity to sum k terms at each time step, which results in an overall computational complexity of $\mathcal{O}(n^2)$ for n time steps.

The hybrid scheme improves the Euler approach by treating the singular part of the kernel with a dedicated quadrature method while using the Euler discretization for the remainder, as given

by Bennedsen et al. (2017). The original hybrid scheme is designed for the generalized Brownian semistationary process

$$Y_t = \int_{-\infty}^{t} K(t-s)\sigma_s \, dW_s,$$

where σ_s is a covariance-stationary process satisfying $\mathbb{E}[\sigma_s] = \mathbb{E}[\sigma_t]$, $\mathrm{Cov}(\sigma_s, \sigma_t) = \mathrm{Cov}(\sigma_0, \sigma_{|s-t|})$ for $s, t \in \mathbb{R}$, and the kernel $K(\cdot)$ exhibits a power-law behavior near zero, i.e.,

$$K(u) = u^{\alpha} L_K(u), \quad u \to 0^+,$$

with $\alpha \in (-\frac{1}{2}, \frac{1}{2}) \setminus \{0\}$ and L_K a slowly varying function. In the case of rough Heston, $\alpha = H - 1/2$, $L_K(u) = \frac{1}{\Gamma(H+1/2)}$. This singularity leads to rough sample paths and renders the Euler-type discretization inaccurate, as it fails to capture the steep behavior of $K(u)$ for small u.

For a discretization grid $\{t, t-\Delta t, t-2\Delta t, \cdots\}$ with time step $\Delta t = 1/n$, the approximation of Y_t is split into two parts:

- **Singular part:** For the κ most recent intervals (where the kernel is most singular), the kernel is approximated by its power-law form and integrated analytically. For example, on the interval $[t-k\Delta t, t-(k-1)\Delta t]$, one approximates

$$K(t-s) \approx L_K(k\Delta t)(t-s)^{\alpha}, \quad k = 1, \ldots, \kappa.$$

- **Regular part:** For larger lags ($k > \kappa$), the kernel is approximated by a constant on each interval using an evaluation point $b_k \in [k-1, k]$, so that

$$K(t-s) \approx K\Big(b_k \Delta t\Big).$$

The resulting discretized process is expressed as

$$Y_t^n = \underbrace{\sum_{k=1}^{\kappa} L_K(k/n)\, \sigma_{t-k/n} \int_{t-k/n}^{t-(k-1)/n} (t-s)^{\alpha} dW_s}_{\text{Singular part}} + \underbrace{\sum_{k=\kappa+1}^{N_n} K(b_k/n)\, \sigma_{t-k/n} \Big(W_{t-(k-1)/n} - W_{t-k/n}\Big)}_{\text{Regular part}}, \tag{10}$$

where N_n is chosen to ensure the tail of the integral is negligible.

The main theoretical result regarding the mean square error of the hybrid scheme is given by the following proposition.

Proposition 2 (Theorem 2.5, Bennedsen et al. (2017))**.** *Suppose that the kernel K satisfies Assumption 1, σ_t satisfies covariance–stationarity*

$$\mathbb{E}|\sigma_s - \sigma_0|^2 = \mathcal{O}(s^{2\alpha+1+\delta}) \quad \text{as } s \downarrow 0,$$

for some $\delta > 0$, and the truncation index N_n satisfies $N_n \sim n^{\gamma+1}$ with

$$\gamma > \frac{-2\alpha + 1}{2\beta + 1},$$

then for all $t \in \mathbb{R}$,

$$\mathbb{E}|Y_t - Y_t^n|^2 \sim J(\alpha, \kappa, \mathbf{b})\, \mathbb{E}[\sigma_0^2]\, n^{-(2\alpha+1)} \{L_K(1/n)\}^2, \quad n \to \infty,$$

where the constant

$$J(\alpha, \kappa, \mathbf{b}) := \sum_{k=\kappa+1}^{\infty} \int_{k-1}^{k} \left(y^\alpha - b_k^\alpha\right)^2 dy < \infty,$$

and $\mathbf{b} = (b_k)_{k=\kappa+1}^{\infty}$ is the chosen sequence of evaluation points with $b_k \in [k-1, k] \setminus \{0\}$.

For kernel function $K(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}$, the above result yields a strong order of H. If the volatility process σ_t is known, the time complexity of the original hybrid scheme can be reduced to $\mathcal{O}(N_n \log N_n)$ using the fast Fourier transform. However, these properties are not directly applicable to our models of interest, since the volatility process σ_t is endogenously determined by $\{Y_u : u \le t\}$, pushing the time complexity back to $\mathcal{O}(N_n^2)$; and the process (4) and (7) are not necessarily covariance-stationary, which baffles the mean square error analysis in Proposition 2.

3.2 Markovian approximation methods

The Markovian approximation method, as discussed in Abi Jaber (2019), Abi Jaber and El Euch (2019), Alfonsi and Kebaier (2023), and Bayer and Breneis (2023), seeks to transform the non-Markovian structure of rough volatility models into a higher-dimensional Markovian framework. The key idea is to use a Fourier–Laplace transform to represent the fractional kernel $K(\cdot)$, which is typically of the form

$$K(t) = \int_0^\infty e^{-xt} \mu(dx) \approx \sum_{j=1}^{n_f} w_j e^{-x_j t}, \tag{11}$$

where $\mu(dx) \approx \sum_{j=1}^{n_f} w_j\, \delta_{x_j}(dx)$. For $K(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}$, $\mu(dx) = \frac{x^{-H-1/2}}{\Gamma(1/2-H)\Gamma(1/2+H)} dx$. Replacing the kernel in process (7) with (11), and applying the stochastic Fubini theorem, one obtains a Markovian approximation of the form

$$X_t = X_0 + \int_0^t \int_0^\infty e^{-x(t-s)} \mu(dx) \left(b(X_s)\, ds + \sigma(X_s)\, dW_s\right) \approx X_0 + \sum_{j=1}^{n_f} w_j Z_t^j, \tag{12}$$

where

$$Z_t^j = \int_0^t e^{-x_j(t-s)} \left(b(X_s)\, ds + \sigma(X_s)\, dW_s\right).$$

It is obvious that Z_t^j satisfies the following SDE

$$dZ_t^j = \left(-x_j Z_t^j + b(X_t)\right) dt + \sigma(X_t)\, dW_t, \quad Z_0^j = 0, \tag{13}$$

which is a Markovian process. The stochastic Volterra equation (4) can thus be approximated by a Markovian process of the form

$$X_t^{n_f} = X_0 + \sum_{j=1}^{n_f} w_j Z_t^j, \tag{14}$$

$$dZ_t^j = (-x_j Z_t^j + b(X_t^{n_f}))\, dt + \sigma(X_t^{n_f})\, dW_t, \quad Z_0^j = 0, \quad j = 1, ..., n_f, \tag{15}$$

where n_f is the number of factors used to approximate the Fourier–Laplace transform (11).

Likewise, the rough Heston model (6)-(7) can be transformed into a similar higher-dimensional Markovian process (or lifted Heston model), given by

$$dS_t^{n_f} = \sqrt{V_t^{n_f}} S_t^{n_f}\, dB_t, \tag{16}$$

$$V_t^{n_f} = g_0(t) + \sum_{j=1}^{n_f} w_j U_t^j, \quad g_0(t) = V_0 + \frac{\lambda\theta}{\Gamma(H+1/2)} \int_0^t (t-s)^{H-1/2} ds \tag{17}$$

$$dU_t^j = (-x_j U_t^j - \lambda V_t^{n_f})\, dt + \nu \sqrt{V_t^{n_f}}\, dW_t, \quad U_0^j = 0, \quad j = 1, ..., n_f, \tag{18}$$

where B, W are defined in the same way as before. Though the characteristic function of $(\log S_T^{n_f}, V_T^{n_f})$ is tractable, the computational efforts required to obtain the exact solution are still substantial (see Appendix B, Abi Jaber (2019)). Instead, it is recommended to simulate the lifted Heston model (16)-(18) using the Euler-type scheme, e.g.,

$$\widehat{V}_{t_k}^{n_f} = g_0(t_k) + \sum_{j=1}^{n_f} w_j\, \widehat{U}_{t_k}^j, \tag{19}$$

$$\widehat{U}_{t_{k+1}}^j = \frac{1}{1 + x_j\, \Delta t} \left(\widehat{U}_{t_k}^j - \lambda\, \widehat{V}_{t_k}^{n_f}\, \Delta t + \nu \sqrt{(\widehat{V}_{t_k}^{n_f})_+}\, W_k^n \right), \quad j = 1, \ldots, n_f, \quad \widehat{U}_0^j = 0, \tag{20}$$

on the discrete-time grid $\{t_k = k\Delta t : k = 0, 1, \ldots, n\}$, where $\Delta t = T/n$, and $W_k^n \sim \mathcal{N}(0, \Delta t)$. The associated time complexity is $\mathcal{O}(n_f \cdot n)$, which is more efficient than the Euler-type scheme for the original rough Heston model.

Convergence result. The convergence result for the Markovian approximation method critically depends on the choice of the parameters $\{(w_j, x_j)\}_{j=1}^{n_f}$, as suggested by following propositions.

Proposition 3 (Theorem 3.1, Alfonsi and Kebaier (2023))**.** *Let X_t be the multidimensional stochastic Volterra equation of the form (1) where $X_0 \in \mathbb{R}^d$, $b : \mathbb{R}^d \to \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \to \mathcal{M}_d(\mathbb{R})$ are globally Lipschitz continuous functions, W is a standard Brownian motion in $\mathbb{R}^d$, and K is a kernel satisfying assumptions in Proposition 2. Let $X_t^{n_f}$ be the Markovian approximation of X_t by replacing kernel K with $\widehat{K}(t) = \sum_{j=1}^{n_f} w_j e^{-x_j t}$. Then, there exists a constant $C > 0$ (depending on $|X_0|, b, \sigma, K$) such that*

$$\mathbb{E}\left|X_t - X_t^{n_f}\right|^2 \leq C \left(\int_0^t |K(s) - \widehat{K}(s)|^2 ds \right), \quad \forall t \in [0, T]. \tag{21}$$

Even if the coefficients b and σ do not satisfy the Lipschitz continuity (Assumption 2(i)), the Markovian approximation $X_t^{n_f}$ still converges to the true process X_t as $n_f \to \infty$, provided that the kernel approximation $\widehat{K}$ satisfies the Assumption 3. The convergence property without the Lipschitz continuity assumption is established in the following proposition.

Proposition 4 (Theorem 7.8, Abi Jaber (2018))**.** *Let $X = (X_t)_{t\geq 0}$ be the stochastic Volterra equation of the form* (1) *with coefficients b and σ satisfying the linear growth condition (Assumption 2(ii)), and the sequence $X^{n_f} = (X_t^{n_f})_{t\geq 0}$ be the continuous weak solutions to (1) with the kernel $\widehat{K}$ satisfying Assumptions 1 and 3. Further assume that*

$$\int_0^T |K(s) - \widehat{K}(s)|^2 \, ds \to \infty, \quad n_f \to \infty,$$

for any given $T > 0$. Then, any point limit of X^{n_f} as $n_f \to \infty$ is a solution of the stochastic Volterra equation (1) *with kernel K.*

Based on Propositions 3 and 4, the basic strategy is to select the parameters $\{(w_j, x_j)\}_{j=1}^{n_f}$ such that

$$\|K - \widehat{K}\|_{2,T} \to 0 \quad \text{as} \quad n_f \to \infty.$$

In accordance with this requirement, various choices for $\{(w_j, x_j)\}_{j=1}^{n_f}$ have been proposed in the literature, each offering distinct convergence properties. We summarize some classical examples below.

Lifted Heston model (Abi Jaber (2018)). In this case,$\{(w_j, x_j)\}_{j=1}^{n_f}$ are chosen such that

$$w_j = \frac{(r_{n_f}^{1-\alpha} - 1) r_{n_f}^{(\alpha-1)(1+n_f/2)}}{\Gamma(\alpha)\Gamma(2-\alpha)} r_{n_f}^{(1-\alpha)j}, \quad x_j = \frac{(1-\alpha)(r_{n_f}^{2-\alpha} - 1)}{(2-\alpha)(r_{n_f}^{1-\alpha} - 1)} r_{n_f}^{j-1-n_f/2}, \; j = 1, \ldots, n_f \tag{22}$$

with $r_{n_f} = 1 + 10\, n_f^{-0.9}$ for $n_f \geq 1$, and $\alpha = H + 1/2$ for $H \in (0, 1/2)$.

For kernel $K(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}$, this choice of parameters leads to

$$\begin{aligned} \|K - \widehat{K}\|_{2,T} \leq C \frac{T^{5/2}}{\sqrt{5}} r_{n_f}^{1/2} (r_{n_f} - 1)^2 \int_0^\infty (1 \wedge x^{-1/2}) \mu(dx) \\ + \frac{r_{n_f}^{-(1-\alpha)n_f/2}}{\Gamma(\alpha)\Gamma(2-\alpha)} + \frac{r_{n_f}^{(1/2-\alpha)n_f/2}}{\Gamma(\alpha)\Gamma(1-\alpha)(1/2-\alpha)} \to 0, \quad n_f \to \infty, \end{aligned} \tag{23}$$

where C is a constant, and $\mu(dx) = \frac{x^{-H-1/2}}{\Gamma(1/2-H)\Gamma(1/2+H)} dx$ (Lemma 8.5, Abi Jaber (2018)). The time complexity of generating this parameter set is $\mathcal{O}(n_f)$.

Gaussian quadrature rule (Bayer and Breneis (2023)). This approach partitions the domain of integration $[0,\infty)$ in (11) into n disjoint subintervals $[\xi_{j-1},\xi_j]$ for $j=1,\dots,n$, where $mn=n_f$. These subintervals are determined by a set of parameters (α,β,a,b) together with $A:=(\frac{1}{H}+\frac{1}{3/2-H})^{1/2}$, such that

$$m\approx\frac{\beta}{A}\sqrt{n_f},\quad n\approx\frac{A}{\beta}\sqrt{n_f},$$

$$\xi_0=a\exp\left(-\frac{\alpha}{(3/2-H)A}\sqrt{n_f}\right),\quad \xi_n=b\exp\left(\frac{\alpha}{HA}\sqrt{n_f}\right),\quad \xi_i=\xi_0\left(\frac{\xi_n}{\xi_0}\right)^{i/n},\quad i=1,\dots,n,$$

On each subinterval $[\xi_{j-1},\xi_j]$, $\{(w_i,x_i)\}_{i=(j-1)m+1}^{jm}$ are determined by the Gaussian quadrature rule, which ensures that $\sum_{i=(j-1)m+1}^{jm} w_i e^{-x_i t}$ is a good approximation of $\int_{\xi_{j-1}}^{\xi_j} e^{-xt}\mu(dx)$. This is done by recurrently finding the orthogonal polynomial p_m of degree m by

$$p_{r+1}(x)=\left(x-\frac{\langle xp_r,p_r\rangle}{\langle p_r,p_r\rangle}\right)p_r(x)-\sum_{j=0}^{r-1}\frac{\langle xp_r,p_j\rangle}{\langle p_j,p_j\rangle}p_j(x)$$

starting from $p_0(x)=1$, where $\langle f,g\rangle=\int_{\xi_{j-1}}^{\xi_j} f(x)g(x)\mu(dx)$. The parameters x_i are then determined by the roots of p_m, i.e., $p_m(x_i)=0$, and the weights w_i are given by

$$w_i=\frac{\langle p_{m-1},p_{m-1}\rangle}{p_m'(x_i)p_{m-1}(x_i)}.$$

In addition, this method sets $x_0=0$ and $w_0=\int_0^{\xi_0}\mu(dx)=\frac{c_H}{1/2-H}\xi_0^{1/2-H}$, where $c_H=\frac{1}{\Gamma(1/2-H)\Gamma(1/2+H)}$. The kernel $K(t)=\frac{t^{H-1/2}}{\Gamma(H+1/2)}$ is then approximated by $\sum_{i=0}^{mn} w_i e^{-x_i t}$.

Generally, the time complexity of generating (w_i,x_i) for one subinterval is $\mathcal{O}(m^2)$, and the overall time complexity of generating the full parameter set is $\mathcal{O}(nm^2)$. The convergence rate of this choice of parameters is superpolynomial with respect to n_f, as shown in Theorem 2.1, Bayer and Breneis (2023): given $\alpha=1.06418,\ \beta=0.4275,\ a=b=1$,

$$\mathbb{E}\left|X_T-X_T^{n_f}\right|\le Cc_H^2\left(\frac{T^3}{(3/2-H)^2}+\frac{3}{2H^2}+\frac{5\pi^3}{48}\left(e^{\alpha\beta}-1\right)^2\frac{A^{2-2H}T^{2H}}{\beta^{2-2H}H}N^{1-H}\right)\exp\left(-\frac{2\alpha}{A}\sqrt{N}\right),$$

where $C, X_T, X_T^{n_f}$ are defined as in Proposition 3.

The Markovian approximation method is not without limitations. The analytic tractability of the characteristic function hinges on solving an n_f-dimensional system of Riccati ordinary differential equations, which still remains computationally intensive. While the fast simulation of the Markovian approximation can be achieved with a complexity of $\mathcal{O}(n_f\cdot n)$ (with n denoting the number of time steps), the accuracy of the method is also dependent on the discretization scheme used. A direct Euler-type approximation applied to the lifted system may suffer from numerical instability.

4 Our method: Accelerated Hybrid Scheme

Based on the limitations of existing discretization methods and Markovian approximations, we develop an Accelerated Hybrid Scheme for rough volatility models. While the hybrid scheme in Bennedsen et al. (2017) offers accurate approximations for singular kernels, its computational complexity becomes prohibitive when applied to models of the type (4) where the process X_t depends on its entire history. Similarly, simulations based on Markovian approximation have their own challenges including slow convergence with respect to the increasing number of factors and potential numerical instability.

Our Accelerated Hybrid Scheme addresses these limitations by introducing a precomputation step that enables the fast Fourier transform to approximate the variance process in the hybrid scheme, which reduces the $\mathcal{O}(n^2)$ time complexity down to $\mathcal{O}(n \log n)$ (n is the number of simulation steps).

The choice of precomputation technique is crucial. To balance the trade-off between computational complexity and accuracy, we expect the precomputation to finish within time linear in n. A feasible choice is to leverage the simulation of Markovian approximation (17)-(18), as it is computationally efficient with a time complexity of $\mathcal{O}(n_f \cdot n)$. Compared to a direct simulation of the approximated process, following it with a fast Fourier hybrid scheme increases the precision and numerical stability.

Notations. In this section, we denote by S_t, V_t, and X_t the processes defined in (6), (7), and (4), respectively, and by U_t^j the j-th auxiliary process in the lifted Heston model (18). Moreover, $S_t^{n_f}$, $V_t^{n_f}$, and $X_t^{n_f}$ represent the Markovian approximations for a rough process with kernel K approximated by n_f factors; $\widehat{S}_t^{n_f}$, $\widehat{V}_t^{n_f}$, $\widehat{X}_t^{n_f}$, and $\widehat{U}_t^j$ denote the Euler-discretized Markovian approximation processes (e.g., as in (16)-(18)); and $\widetilde{S}_t$, $\widetilde{V}_t$, $\widetilde{X}_t$ indicate the corresponding simulated processes in the Accelerated Hybrid Scheme. n typically refers to the number of time steps employed in the simulation.

4.1 Algorithm details

In this subsection, we first investigate the simulation of the general stochastic Volterra equation

$$X_t = X_0 + \int_0^t K(t-s)b(X_s)\,ds + \int_0^t K(t-s)\sigma(X_s)\,dW_s,$$

as presented in (4) with kernel function $K(u) = \frac{u^{H-1/2}}{\Gamma(H+1/2)}$, $H \in (0, \frac{1}{2})$. Subsequently, we extend our method to the rough Heston model given by (6)-(7).

4.1.1 Simulation of X_t

The simulation algorithm starts from the Markovian approximation of X_t, which is given by

$$X_t^{n_f} = X_0 + \sum_{j=1}^{n_f} w_j Z_t^j, \tag{24}$$

$$dZ_t^j = (-x_j Z_t^j + b(X_t^{n_f}))\, dt + \sigma(X_t^{n_f})\, dW_t, \quad Z_0^j = 0, \quad j = 1, ..., n_f. \tag{25}$$

To balance between convergence rate and computational complexity, our algorithm chooses $\{(w_j, x_j)\}_{j=1}^{n_f}$ according to (22), which is originally given by Abi Jaber (2018). The rest of the algorithm is outlined as follows.

Precomputation. First, we simulate the Markovian approximation (24)-(25) with n_f factors on the discrete-time grid $\pi_n = \{0, \frac{T}{n}, \frac{2T}{n}, ..., T\}$ where $t_i = \frac{iT}{n}$. The simulation is performed using the Euler-type scheme given by

$$\widehat{X}_{t_k}^{n_f} = X_0 + \sum_{j=1}^{n_f} w_j \widehat{Z}_{t_k}^j, \tag{26}$$

$$\widehat{Z}_{t_{k+1}}^j = e^{-x_j \Delta t} \left(\widehat{Z}_{t_k}^j + b(\widehat{X}_{t_k}^{n_f})\Delta t + \sigma(\widehat{X}_{t_k}^{n_f})\Delta W_k \right), \quad \widehat{Z}_0^j = 0, \quad j = 1, \ldots, n_f, \tag{27}$$

where $\Delta W_k \sim \mathcal{N}(0, \Delta t),\ k = 0, 1, ..., n-1,\ \Delta t := T/n$. The time complexity of this step is $\mathcal{O}(n_f \cdot n)$.

Hybrid approximation. Building upon the precomputed values $\{\widehat{X}_t^{n_f} : t \in \pi_n\}$, we define the accelerated hybrid approximation $\widetilde{X}_{t_k}$ of X_{t_k} on the discrete-time grid π_n by

$$\begin{aligned} \widetilde{X}_{t_k} = X_0 &+ \sum_{i=1}^{k} b\big(\widehat{X}_{t_{k-i}}^{n_f}\big) \int_{(k-i)T/n}^{(k-i+1)T/n} K(t_k - s)\, ds \\ &+ \sum_{i=1}^{\kappa} \sigma\big(\widehat{X}_{t_{k-i}}^{n_f}\big) \int_{(k-i)T/n}^{(k-i+1)T/n} K(t_k - s)\, dW_s + \sum_{i=\kappa+1}^{k} K\Big(\frac{b_i T}{n}\Big)\, \sigma\big(\widehat{X}_{t_{k-i}}^{n_f}\big)\Delta W_{k-i}, \end{aligned} \tag{28}$$

where ΔW_k denotes the Brownian increment over the interval $[kT/n, (k+1)T/n]$ (i.e. $\Delta W_k \sim \mathcal{N}(0, T/n)$), and

$$W_{k-i,i} := \int_{(k-i)T/n}^{(k-i+1)T/n} (t_k - s)^{H-1/2}\, dW_s, \quad i = 1, \ldots, \kappa,$$

can be simulated exactly via a multivariate Gaussian distribution. The parameter sequence $b_i \in [i-1, i]$ is selected to yield an adequate approximation for $K(u)$ on $u \in [(i-1)T/n,\ iT/n]$. Here, we choose

$$b_i = \left(\frac{i^{H+1/2} - (i-1)^{H+1/2}}{H + 1/2} \right)^{\frac{1}{H-1/2}} \tag{29}$$

so that the mean square error of the approximation is minimized (see Theorem 1).

For kernel $K(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}$, we have

$$\int_{(k-i)T/n}^{(k-i+1)T/n} K(t_k - s)\, ds = \frac{i^{H+1/2} - (i-1)^{H+1/2}}{H+1/2} \left(\frac{T}{n}\right)^{H+1/2} = K\Big(\frac{b_i T}{n}\Big)\frac{T}{n}.$$

Thus, the sequence $\widetilde{X}_{t_1}, \widetilde{X}_{t_2}, \ldots, \widetilde{X}_{t_n}$ can be computed efficiently as follows.

In (28), we let $K(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}$, and denote

$$\mathcal{I}_k = \sum_{i=1}^{\kappa} b\big(\widehat{X}^{n_f}_{t_{k-i}}\big) K\left(\frac{b_i T}{n}\right)\frac{T}{n} + c_H \sum_{i=1}^{\kappa} \sigma\big(\widehat{X}^{n_f}_{t_{k-i}}\big) W_{k-i,i}, \tag{30}$$

$$\mathcal{J}_k = \sum_{i=\kappa+1}^{k} b\big(\widehat{X}^{n_f}_{t_{k-i}}\big) K\left(\frac{b_i T}{n}\right)\frac{T}{n} + \sum_{i=\kappa+1}^{k} K\Big(\frac{b_i T}{n}\Big)\, \sigma\big(\widehat{X}^{n_f}_{t_{k-i}}\big)\Delta W_{k-i}, \quad k = 1, ..., n, \tag{31}$$

where $c_H = \frac{1}{\Gamma(H+1/2)}$.

The computation of (28) is then partitioned into two components. The first, namely the evaluation of $\mathcal{I}_k$, involves summing only κ terms ($\kappa \ll n$) for each $\widetilde{X}_{t_k}$ and is thus highly efficient. In contrast, the evaluation of $\mathcal{J}_k$ entails a discrete convolution which, if computed by brute force, would require $\mathcal{O}(n^2)$ operations. By employing the fast Fourier transform, this convolution can be computed in $\mathcal{O}(n \log n)$ time.

We define the discrete convolution as follows:

$$\Gamma_k = \begin{cases} 0, & k = 1, \ldots, \kappa, \\ K\Big(\frac{b_k T}{n}\Big), & k = \kappa+1, \ldots, n, \end{cases}$$

and

$$\Xi_k = b\big(\widehat{X}^{n_f}_{t_{k-1}}\big)\frac{T}{n} + \sigma\big(\widehat{X}^{n_f}_{t_{k-1}}\big)\Delta W_{k-1}, \quad k = 1, \ldots, n.$$

Accordingly, we express

$$\mathcal{J}_i = \sum_{k=1}^{i} \Gamma_k \Xi_{i+1-k}, \quad i = 1, \ldots, n, \tag{32}$$

which enables computing the sequence $\mathcal{J}_1, \ldots, \mathcal{J}_n$ via FFT with complexity $\mathcal{O}(n \log n)$.

The detailed algorithm for the simulation of X_t is summarized in Algorithm 1.

4.1.2 Simulation of the rough Heston model

The Accelerated Hybrid Scheme for the rough Heston model (6)–(7) is derived as a specific instance of Algorithm 1 by selecting

$$b(x) = \lambda\big(\theta - x\big) \quad \text{and} \quad \sigma(x) = \nu\sqrt{x}\,.$$

Algorithm 1 Simulation of X_t in (4) using Accelerated Hybrid Scheme

Input: X_0, $b(\cdot)$, $\sigma(\cdot)$, H, n_f, n, T, κ
$t_i \leftarrow \frac{iT}{n}, \quad \widehat{X}^{n_f}_{t_i} \leftarrow X_0, \quad i = 0, 1, \ldots, n, \quad \Delta t \leftarrow T/n$
$\widehat{Z}^j_{t_0} \leftarrow 0, \quad j = 1, \ldots, n_f$
Generate $\mathbf{W}_k = (\Delta W_k, W_{k,1}, \ldots, W_{k,\kappa}),\ k = 0, 1, ..., n-1$ according to Bennedsen et al. (2017)
Compute $\{(w_i, x_i)\}_{i=1}^{n_f}$ using (22)
for $k = 0, 1, \ldots, n$ **do**
  $\widehat{X}^{n_f}_{t_k} \leftarrow \widehat{X}^{n_f}_{t_k} + \sum_{j=1}^{n_f} w_j \widehat{Z}^j_{t_k}$
  for $i = 1, \ldots, n_f$ **do**
    $\widehat{Z}^j_{t_{k+1}} = e^{-x_j \Delta t} \left(\widehat{Z}^j_{t_k} + b(\widehat{X}^{n_f}_{t_k})\Delta t + \sigma(\widehat{X}^{n_f}_{t_k})\Delta W_k \right)$
  end for
end for
Compute $\mathcal{J}_1, \mathcal{J}_2, \ldots, \mathcal{J}_n$ using FFT as defined in (32)
for $k = 1, ..., n$ **do**
  $\widetilde{X}_{t_k} \leftarrow X_0 + \mathcal{J}_k$
  for $j = 1, ..., \kappa$ **do**
    if $k - j \geq 0$ **then**
      $\widetilde{X}_{t_k} \leftarrow \widetilde{X}_{t_k} + \frac{1}{\Gamma(H+1/2)} \left(b(\widehat{X}^{n_f}_{t_{k-j}}) \left(\frac{b_j T}{n} \right)^{H-1/2} \frac{T}{n} + \sigma(\widehat{X}^{n_f}_{t_{k-j}}) W_{k-j,j} \right)$
    end if
  end for
end for
Output: $\{\widetilde{X}_{t_k} : k = 1, ..., n\}$

In this context, the variance process V_t is approximated via the Markovian approximation (19)-(20), and the Accelerated Hybrid Scheme is applied to V_t analogously to X_t. In addition, the deterministic component

$$g_0(t) = V_0 + \frac{1}{\Gamma(H + 1/2)} \int_0^t (t - s)^{H-1/2} \lambda \theta \, ds$$

is evaluated exactly.

In terms of the fast Fourier transform, we follow the same procedure as in Section 4.1.1 and define the discrete convolution as follows:

$$\mathcal{I}_k = \frac{1}{\Gamma(H + 1/2)} \left(-\sum_{i=1}^{\kappa} \lambda \widehat{V}^{n_f}_{t_{k-i}} \left(\frac{b_i T}{n} \right)^{H-1/2} \frac{T}{n} + \sum_{i=1}^{\kappa} \sqrt{(\widehat{V}^{n_f}_{t_{k-i}})_+} W_{k-i,i} \right), \tag{33}$$

$$\mathcal{J}_k = \sum_{i=1}^{k} \Gamma_i \Xi_{k-i+1}, \tag{34}$$

where

$$\Gamma_i = \begin{cases} 0, & i = 1, \dots, \kappa, \\ \frac{1}{\Gamma(H+1/2)} \left(\frac{b_i T}{n}\right)^{H-1/2}, & i = \kappa+1, \dots, k, \end{cases}$$

$$\Xi_i = -\frac{\lambda T}{n} \widehat{V}^{n_f}_{t_{i-1}} + \nu \sqrt{(\widehat{V}^{n_f}_{t_{i-1}})_+} \Delta W_{i-1}, \quad i = 1, \dots, k,$$

and b_i takes the form (29). The detailed algorithm for the simulation of the rough Heston model is given in Algorithm 2.

4.2 Convergence analysis

The convergence of the Accelerated Hybrid Scheme is guaranteed by the following theorem.

Theorem 1. *Let $\{X_t : t \geq 0\}$ be the process defined by the stochastic Volterra equation (4) on a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ with kernel $K(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}$ and coefficients b, σ satisfying Assumption 2. Let $\{\widetilde{X}_{t_i} : i = 0, 1, \dots, n,\ t_i = \frac{iT}{n}\}$ be the process generated by Algorithm 1. Then, for any $T > 0$,*

$$\mathbb{E}|X_T - \widetilde{X}_T|^2 \to 0, \quad n_f \to \infty, \quad n \to \infty$$

Proof. See Appendix A. □

Following the proof of Theorem 1, we can find that the mean square error of the Accelerated Hybrid Scheme is bounded by

$$\mathbb{E}|X_T - \widetilde{X}_T|^2 \leq C_1 \|K - \widehat{K}\|^2_{2,T} + C_2 n^{-2H}, \quad \text{as } n_f \to \infty,\ n \to \infty, \tag{35}$$

for some constants $C_1, C_2 > 0$, rendering a strong order of convergence H with respect to the discretization time steps n. With a time complexity $\mathcal{O}(n \cdot (n_f + \log n))$, the Accelerated Hybrid Scheme is a rather efficient and accurate method to simulate the process $\{X_t : t \geq 0\}$.

While the Accelerated Hybrid Scheme can be readily applied to rough volatility processes (3) and (7), the presence of non-Lipschitz coefficients in these models may cause deviations from the results stated in Theorem 1. This is primarily due to the challenges in establishing strong convergence for the scheme (19)-(20). A rigorous treatment of this issue is deferred to future research. In the following section, we empirically demonstrate the efficiency and accuracy of the Accelerated Hybrid Scheme in simulating rough volatility processes under relaxed Lipschitz conditions.

5 Numerical experiments

In this section, we present numerical experiments that test the computational performance of our Accelerated Hybrid Scheme. The experiments are conducted in the context of European call option pricing based on the rough Heston model. According to the non-arbitrage pricing methods, the initial

Algorithm 2 Simulation of rough Heston model (6)-(7) using Accelerated Hybrid Scheme

Input: V_0, S_0 λ, ν, θ, ρ, H, n_f, n, T, κ

$\Delta t \leftarrow T/n$

for $i = 0, 1, \ldots, n$ **do**

  $t_i \leftarrow \frac{iT}{n}$

  $g_0(t_i) = V_0 + \frac{1}{\Gamma(H+1/2)} \int_0^{t_i} (t_i - s)^{H-1/2} \lambda\theta \, ds$

  $\widehat{V}_{t_i}^{n_f} \leftarrow g_0(t_i)$

end for

$\widehat{U}_{t_0}^{j} \leftarrow 0, \quad j = 1, \ldots, n_f$

Generate $\mathbf{W}_k = (\Delta W_k, W_{k,1}, \ldots, W_{k,\kappa}),\ k = 0, 1, ..., n-1$ according to Bennedsen et al. (2017)

Generate $\Delta W_k^{\perp} \sim \mathcal{N}(0, \frac{T}{n})$ for $k = 0, 1, ..., n-1$ independent of $\mathbf{W}_k$

Compute $\{(w_i, x_i)\}_{i=1}^{n_f}$ using (22)

for $k = 0, 1, \ldots, n$ **do**

  $\widehat{V}_{t_k}^{n_f} \leftarrow \widehat{V}_{t_k}^{n_f} + \sum_{j=1}^{n_f} w_j \widehat{U}_{t_k}^{j}$

  for $i = 1, \ldots, n_f$ **do**

    $\widehat{U}_{t_{k+1}}^{i} \leftarrow e^{-x_i \Delta t} \left(\widehat{U}_{t_k}^{i} - \lambda \, \widehat{V}_{t_k}^{n_f} \, (t_{k+1} - t_k) + \nu \sqrt{(\widehat{V}_{t_k}^{n_f})_+} \, \Delta W_k \right)$

  end for

end for

Compute $\mathcal{J}_1, \mathcal{J}_2, \ldots, \mathcal{J}_n$ using FFT as defined in (34)

$\log \widetilde{S}_{t_0} \leftarrow \log S_0$

for $k = 1, ..., n$ **do**

  $\log \widetilde{S}_{t_k} \leftarrow \log \widetilde{S}_{t_{k-1}} - \frac{1}{2}(\widetilde{V}_{t_{k-1}})_+ (t_k - t_{k-1}) + \sqrt{(\widetilde{V}_{t_{k-1}})_+} (\rho \, \Delta W_{k-1} + \sqrt{1-\rho^2} \, \Delta W_{k-1}^{\perp})$

  $\widetilde{V}_{t_k} \leftarrow g_0(t_k) + \mathcal{J}_k$

  for $j = 1, ..., \kappa$ **do**

    if $k - j \geq 0$ **then**

      $\widetilde{V}_{t_k} \leftarrow \widetilde{V}_{t_k} - \frac{\lambda}{\Gamma(H+1/2)} \widehat{V}_{t_{k-j}}^{n_f} \left(\frac{b_j T}{n} \right)^{H-1/2} \frac{T}{n} + \frac{\nu}{\Gamma(H+1/2)} \sqrt{(\widehat{V}_{t_{k-j}}^{n_f})_+} W_{k-j,j}$

    end if

  end for

end for

Output: $\{(\log \widetilde{S}_{t_k}, \widetilde{V}_{t_k}) : k = 1, ..., n\}$

arbitrage-free price of a European call option with maturity T and strike price K is given by the form $\mathbb{E}[e^{-rT}(S_T - K)^+]$, where the expectation $\mathbb{E}$ is taken under the risk-neutral measure. For Monte Carlo simulations of the rough Heston process (6)-(7), including the one described in Algorithm 2, we adopt the discrete-time grid $\pi_n = \{0 = t_0 < t_1 < \cdots < t_n = T\}$ where $t_k = \frac{kT}{n}$, and generate N paths of the process on the grid. Without loss of generality, we set risk-free interest rate $r = 0$ so that the dynamics (6)-(7) correspond with the risk-neutral measure. On top of the European call option pricing, implied volatility curves are calculated to further validate the performance of our Accelerated Hybrid Scheme in industrial applications.

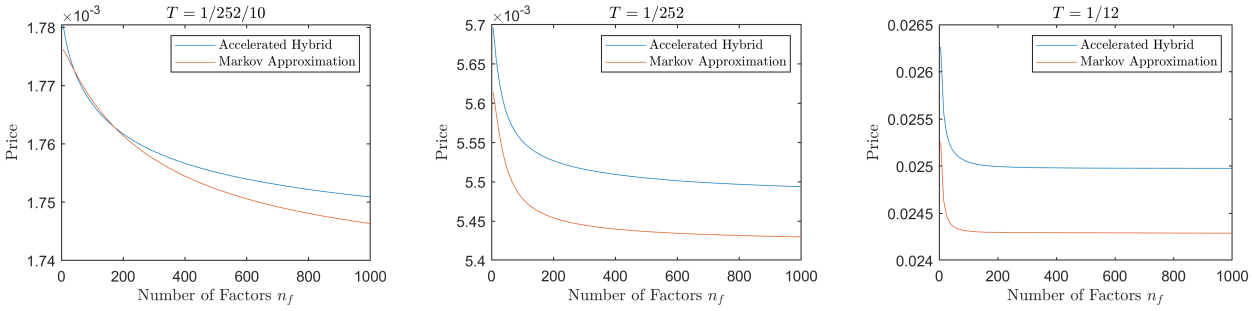
In the following, we first investigate the impact of varying the number of Markovian approximation factors (n_f) and the number of time discretization steps (n) on the accuracy of the scheme. Then, we analyze the behavior of implied volatility surfaces computed using our method. In all experiments, we benchmark our Accelerated Hybrid Scheme against the Fourier transform inversion based pricing method mentioned by Abi Jaber (2018). For the convergence tests, we additionally compare our method with other simulation algorithms including the Euler scheme (9), the implicit-explicit scheme of the Markovian approximation method (19)-(20), and the classical hybrid scheme (10).

5.1 Varying number of approximation factors n_f

First, we examine the European call option pricing results under varying number of approximation factors n_f as a part of our convergence tests. Plots of the estimated option prices against n_f are reported in Figure 1 and 2, indicating a faster convergence of our Accelerated Hybrid Scheme than the simulation of Markovian approximation models as n_f increases in most scenarios. This demonstrates that the FFT-based hybrid scheme effectively mitigates kernel approximation errors introduced during the precomputation step while adding little time cost (the additional CPU time required by our method averages 5.68 seconds across all experiments presented in Figures 1 and 2, measured on a laptop with an Intel Core i7-11800H (2.30GHz) processor and 32 GB RAM, and remains consistent across all values of n_f). Also, it is worth noting that the convergence with respect to n_f is faster for longer maturities and larger Hurst indices. And for Hurst index $H = 0.4$, the pricing results of our method barely change with increasing n_f. These observations are consistent with Proposition 3 and equation (35), in that kernel approximation errors tend to be smaller as H approaches $1/2$.

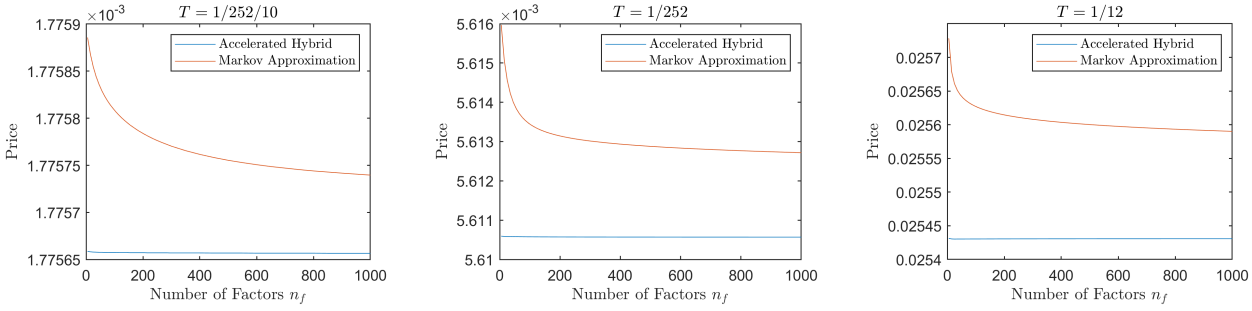
5.2 Varying number of time steps n

Next, we examine the convergence behavior of the Accelerated Hybrid Scheme with respect to the number of time discretization steps n. To isolate the effect of n, we analyze the root mean squared error (RMSE) of the scheme in at-the-money European call option pricing as n_f varies, considering



(a) $T = 1/252/10$ (b) $T = 1/252$ (c) $T = 1/12$

Figure 1: At-the-money European call option prices as the number of Markovian approximation factors n_f increases. The parameters are chosen as $H = 0.1, S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1$, $\kappa = 2$, and $n = 200$.



(a) $T = 1/252/10$ (b) $T = 1/252$ (c) $T = 1/12$

Figure 2: At-the-money European call option prices as the number of Markovian approximation factors n_f increases. The parameters are chosen as $H = 0.4, S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1, n = 200$, $\kappa = 2$, and $n = 200$.

n	Price	RMSE	SE	Bias	CPU time (s)
$n_f = 50$, Reference Price = 0.005404					
25	0.005608	2.044e-04	1.545e-05	2.038e-04	0.67
50	0.005596	1.929e-04	1.543e-05	1.923e-04	1.27
100	0.005602	1.989e-04	1.544e-05	1.983e-04	2.41
250	0.005604	2.010e-04	1.544e-05	2.004e-04	7.86
500	0.005603	2.001e-04	1.545e-05	1.995e-04	19.41
$n_f = 100$, Reference Price = 0.005404					
25	0.005583	1.794e-04	1.539e-05	1.788e-04	0.89
50	0.005568	1.645e-04	1.536e-05	1.637e-04	1.68
100	0.005571	1.676e-04	1.537e-05	1.668e-04	2.89
250	0.005571	1.676e-04	1.536e-05	1.669e-04	6.85
500	0.005569	1.658e-04	1.537e-05	1.651e-04	20.21
$n_f = 200$, Reference Price = 0.005404					
25	0.005567	1.641e-04	1.535e-05	1.634e-04	1.16
50	0.005549	1.460e-04	1.531e-05	1.452e-04	2.19
100	0.005549	1.465e-04	1.531e-05	1.457e-04	4.22
250	0.005547	1.439e-04	1.530e-05	1.431e-04	10.64
500	0.005544	1.409e-04	1.530e-05	1.401e-04	27.66
$n_f = 400$, Reference Price = 0.005404					
25	0.005560	1.565e-04	1.533e-05	1.558e-04	2.30
50	0.005538	1.347e-04	1.528e-05	1.338e-04	4.52
100	0.005535	1.320e-04	1.527e-05	1.311e-04	8.98
250	0.005530	1.271e-04	1.525e-05	1.262e-04	22.32
500	0.005526	1.227e-04	1.525e-05	1.218e-04	50.88
$n_f = 800$, Reference Price = 0.005404					
25	0.005557	1.539e-04	1.532e-05	1.532e-04	4.02
50	0.005532	1.287e-04	1.526e-05	1.278e-04	8.01
100	0.005525	1.226e-04	1.524e-05	1.217e-04	17.93
250	0.005517	1.146e-04	1.522e-05	1.136e-04	42.05
500	0.005512	1.090e-04	1.522e-05	1.080e-04	88.55

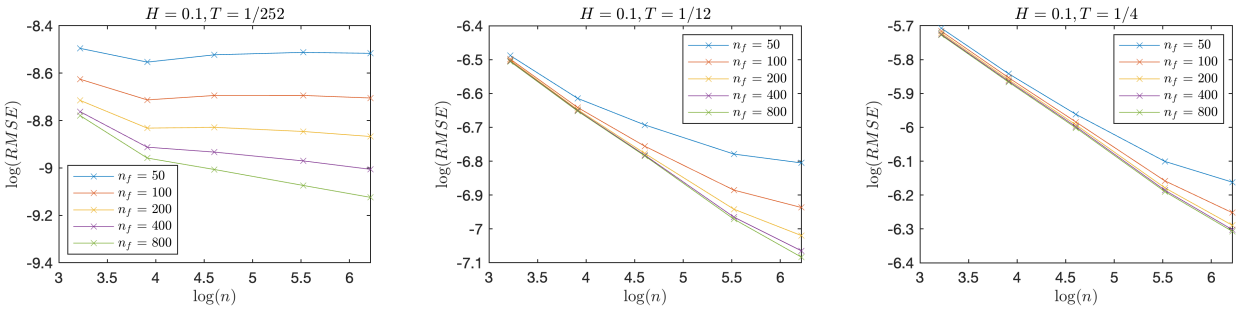
Table 1: Accelerated Hybrid Scheme pricing results of Figure 3(a), $H = 0.1$, $T = 1/252$

Note: Other parameters are set as $S_0 = 1$, $\theta = 0.05$, $\lambda = 0.1$, $\nu = 0.3$, $V_0 = 0.05$, $r = d = 0$, $\rho = -0.1$, and $\kappa = 2$. The reference price is computed using the Fourier transform inversion method. The CPU time is measured on a MacBook Pro with a 10-core Apple M2 Pro processor and 32 GB RAM.

n	Price	RMSE	SE	Bias	CPU time (s)
$n_f = 50$, Reference Price = 0.024095					
25	0.025615	1.521e-03	7.474e-05	1.520e-03	0.60
50	0.025434	1.341e-03	7.429e-05	1.339e-03	1.16
100	0.025333	1.240e-03	7.404e-05	1.237e-03	2.36
250	0.025230	1.137e-03	7.378e-05	1.135e-03	8.25
500	0.025201	1.108e-03	7.369e-05	1.106e-03	21.32
$n_f = 100$, Reference Price = 0.024095					
25	0.025599	1.506e-03	7.470e-05	1.504e-03	0.88
50	0.025399	1.305e-03	7.419e-05	1.303e-03	1.70
100	0.025257	1.164e-03	7.381e-05	1.162e-03	2.92
250	0.025115	1.023e-03	7.345e-05	1.020e-03	6.55
500	0.025064	9.715e-04	7.331e-05	9.688e-04	19.92
$n_f = 200$, Reference Price = 0.024095					
25	0.025594	1.500e-03	7.468e-05	1.499e-03	1.15
50	0.025390	1.297e-03	7.417e-05	1.295e-03	2.16
100	0.025232	1.139e-03	7.374e-05	1.137e-03	4.26
250	0.025059	9.665e-04	7.329e-05	9.637e-04	10.60
500	0.024987	8.943e-04	7.310e-05	8.913e-04	27.82
$n_f = 400$, Reference Price = 0.024095					
25	0.025590	1.497e-03	7.468e-05	1.495e-03	2.32
50	0.025387	1.294e-03	7.416e-05	1.292e-03	4.52
100	0.025226	1.133e-03	7.372e-05	1.131e-03	8.94
250	0.025037	9.442e-04	7.323e-05	9.414e-04	22.34
500	0.024947	8.549e-04	7.299e-05	8.517e-04	51.51
$n_f = 800$, Reference Price = 0.024095					
25	0.025588	1.495e-03	7.467e-05	1.493e-03	4.49
50	0.025385	1.291e-03	7.415e-05	1.289e-03	8.27
100	0.025224	1.131e-03	7.372e-05	1.129e-03	15.94
250	0.025031	9.385e-04	7.322e-05	9.357e-04	39.64
500	0.024931	8.392e-04	7.295e-05	8.360e-04	87.27

Table 2: Accelerated Hybrid Scheme pricing results of Figure 3(b), $H = 0.1$, $T = 1/12$

Note: Other parameters are set as $S_0 = 1$, $\theta = 0.05$, $\lambda = 0.1$, $\nu = 0.3$, $V_0 = 0.05$, $r = d = 0$, $\rho = -0.1$, and $\kappa = 2$. The reference price is computed using the Fourier transform inversion method. The CPU time is measured on a MacBook Pro with a 10-core Apple M2 Pro processor and 32 GB RAM.

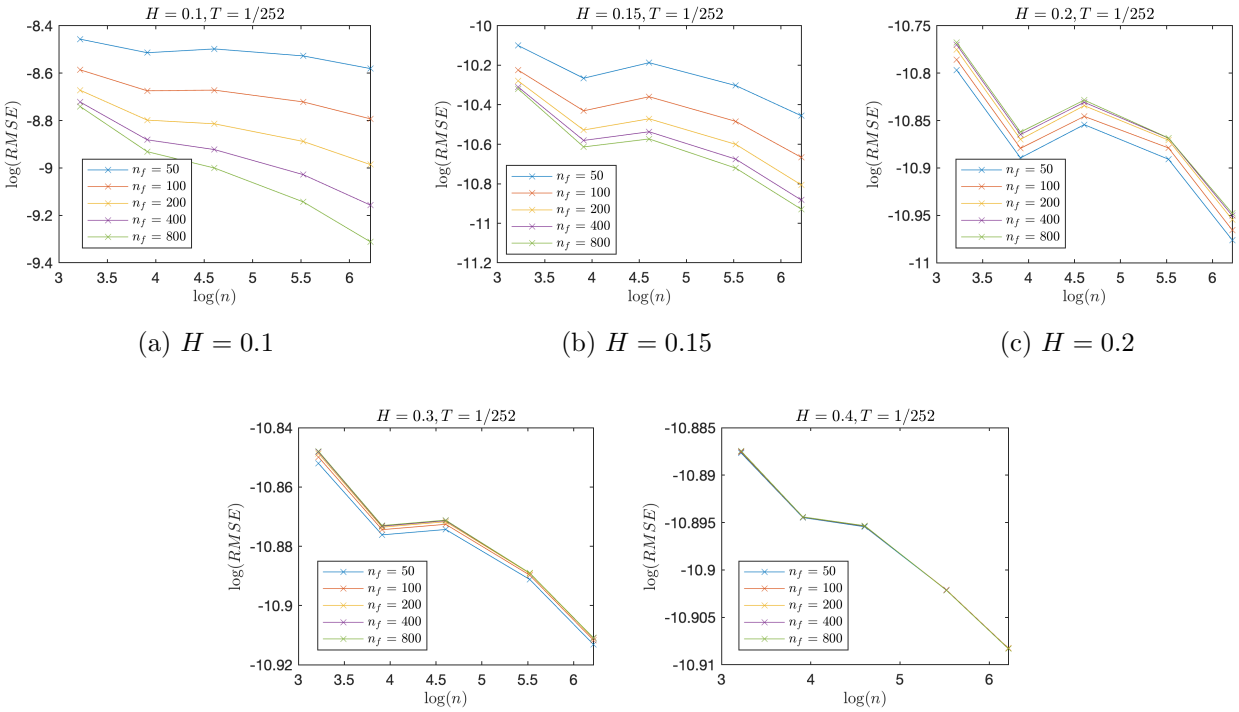


(a) $T = 1/252$ (b) $T = 1/12$ (c) $T = 1/4$

Figure 3: At-the-money European call option pricing errors as the number of discretization time step n increases. RMSE $= \sqrt{\text{standard error}^2 + \text{bias}^2}$. The parameters are chosen as $H = 0.1, S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1$, and $\kappa = 2$.

several representative values of n_f. Under $H = 0.1$, Figure 3 shows that the convergence of the Accelerated Hybrid Scheme tends to be more robust to n_f as the option's time-to-maturity increases, and that sufficient number of Markovian approximation factors in the precomputation step can help the scheme achieve better accuracy. However, the effect of n_f on the convergence is less significant when H becomes larger (the volatility paths become smoother). As shown in Figure 4, the increased number of approximation factors n_f does not necessarily lead to better convergence of the scheme. This is because the kernel approximation error is already small enough when H exceeds a certain value. Further, it deserves noting that the RMSE of the Accelerated Hybrid Scheme may not decrease monotonically as n increases, especially for small T. This is because the bias itself is estimated via Monte Carlo and Fourier transform inversion, thus carrying a numerical error. For very short maturities T, the theoretical discretization bias of the Accelerated Hybrid Scheme becomes negligible compared to the numerical error in the bias estimate, and hence the measured bias fluctuates with n, producing non-monotonic "wobbles" in the total RMSE (see Tables 1 and 2). For longer maturities, the bias stabilizes and the convergence behavior of the RMSE aligns well with the theoretical results in (35), despite discrepancies in model assumptions.

Additionally, we investigate the sensitivity of the Accelerated Hybrid Scheme to the choice of κ, which determines the number of Gaussian random variables used to model the singular part in (10). Notably, when $\kappa = 0$, the FFT-accelerated hybrid scheme reduces to an FFT-accelerated Euler scheme following the precomputation step. This flexibility highlights the adaptability of our framework to different simulation requirements. As shown in Figure 5, while pricing results under $\kappa = 0, 1, 3, 4$ exhibit different convergence behaviors with various choices of H, $\kappa = 2$ demonstrates a balanced performance across the board. This observation supports us to choose $\kappa = 2$ for stable Monte Carlo pricing results. Also, we use $n_f = 400$ for $H < 0.2$ and $n_f = 100$ for $H \geq 0.2$ in the following experiments, as it balances the accuracy and computational efficiency of the Accelerated



(a) $H = 0.1$ (b) $H = 0.15$ (c) $H = 0.2$

(d) $H = 0.3$ (e) $H = 0.4$

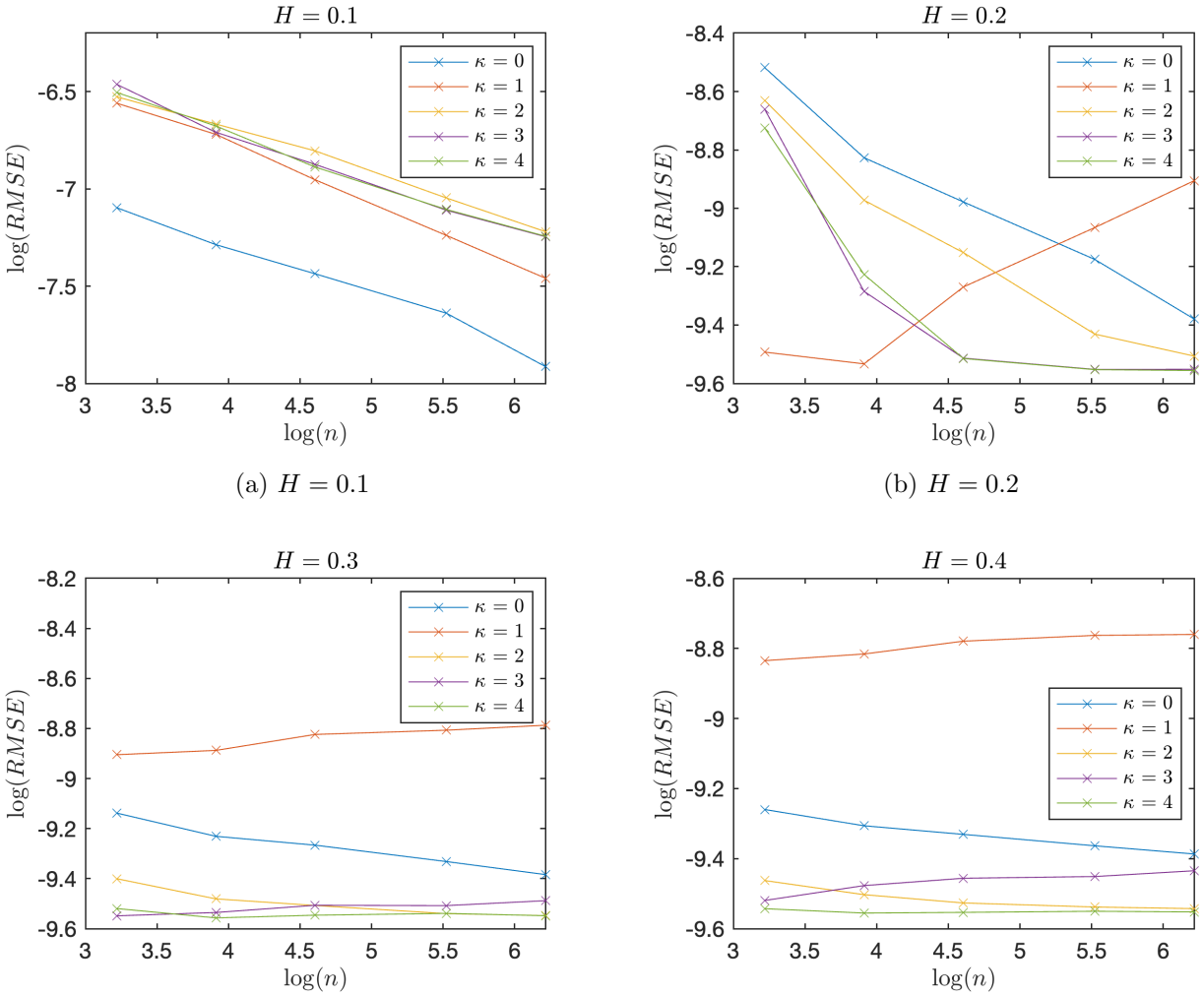
Figure 4: At-the-money European call option pricing errors as the number of discretization time step n increases. RMSE $= \sqrt{\text{standard error}^2 + \text{bias}^2}$. Parameters are chosen as $S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1$, and $\kappa = 2$ along with varying H.

Hybrid Scheme.

Based on the findings above, we compare the convergence behavior of our Accelerated Hybrid Scheme with other simulation algorithms, including the Euler scheme, the implicit-explicit scheme of the Markovian approximation method, and the classical hybrid scheme. The results are summarized in Figure 6 to 10, showing that the Accelerated Hybrid Scheme outperforms the other methods in terms of convergence speed and accuracy when H is not too small and the time-to-maturity T is moderate.

5.3 Applications in implied volatility calculation

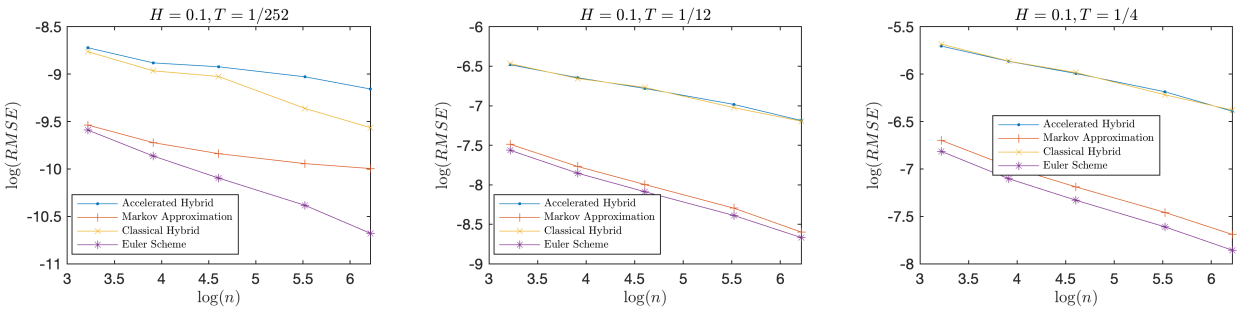
In this subsection, we apply our Accelerated Hybrid Scheme to calculate the implied volatility curves based on European call option pricing results, and compare the computational efficiency and accuracy with other methods. The experiment includes calculating implied volatility curves for 20 consecutive trading days, and deriving an implied volatility surface thereafter. Without loss of generality, we choose $H = 0.15$, $n_f = 400$, and discretization time steps to be $n = 50$ between every two consecutive trading days. Results in Figure 11 indicate that approximation errors of our scheme are all at the



(a) $H = 0.1$ (b) $H = 0.2$

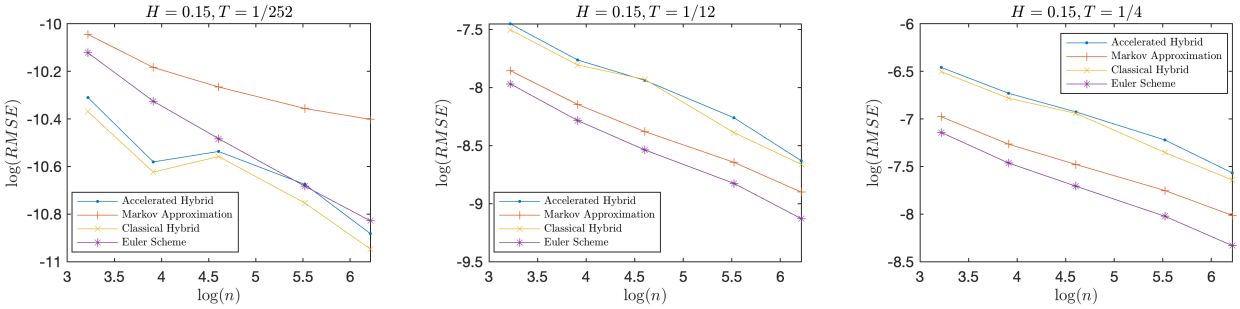
(c) $H = 0.3$ (d) $H = 0.4$

Figure 5: At-the-money European call option pricing errors as the number of discretization time step n increases under varying κ. RMSE $= \sqrt{\text{standard error}^2 + \text{bias}^2}$. Parameters are chosen as $T = 1/12, S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1, n_f = 800$.



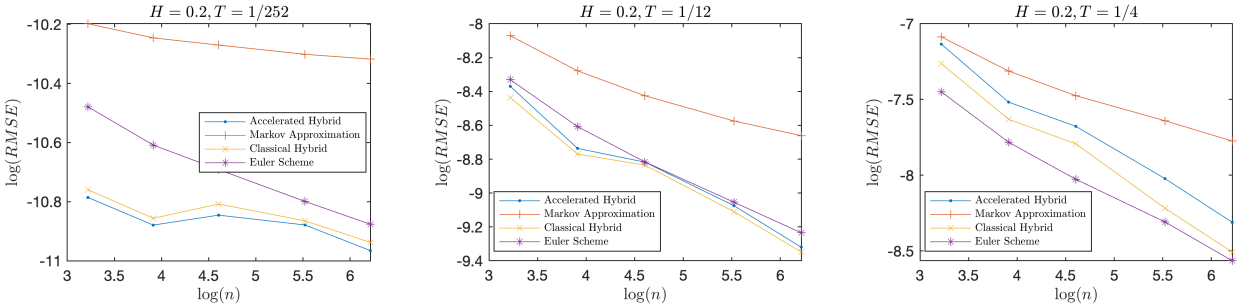
(a) $T = 1/252$ (b) $T = 1/12$ (c) $T = 1/4$

Figure 6: At-the-money European call option pricing errors under $H = 0.1$. RMSE $= \sqrt{\text{standard error}^2 + \text{bias}^2}$. Parameters are chosen as $S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1, \kappa = 2, n_f = 400$.



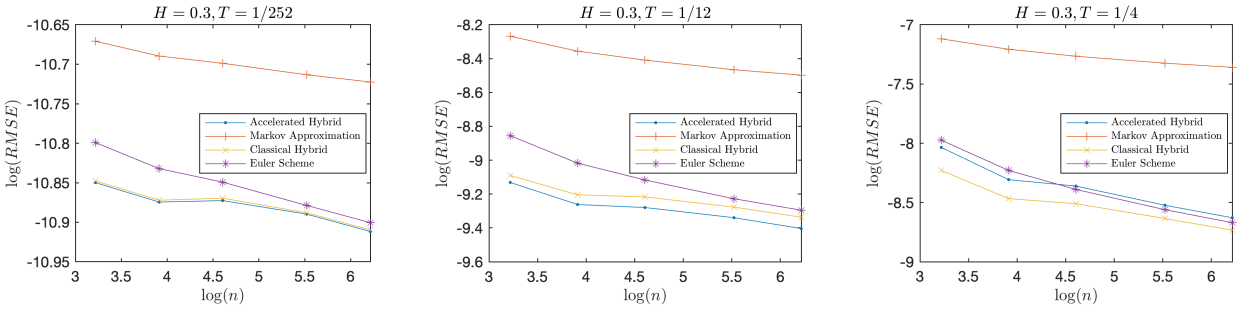
(a) $T = 1/252$ (b) $T = 1/12$ (c) $T = 1/4$

Figure 7: At-the-money European call option pricing errors under $H = 0.15$. RMSE $= \sqrt{\text{standard error}^2 + \text{bias}^2}$. Parameters are chosen as $S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1, \kappa = 2, n_f = 400$.



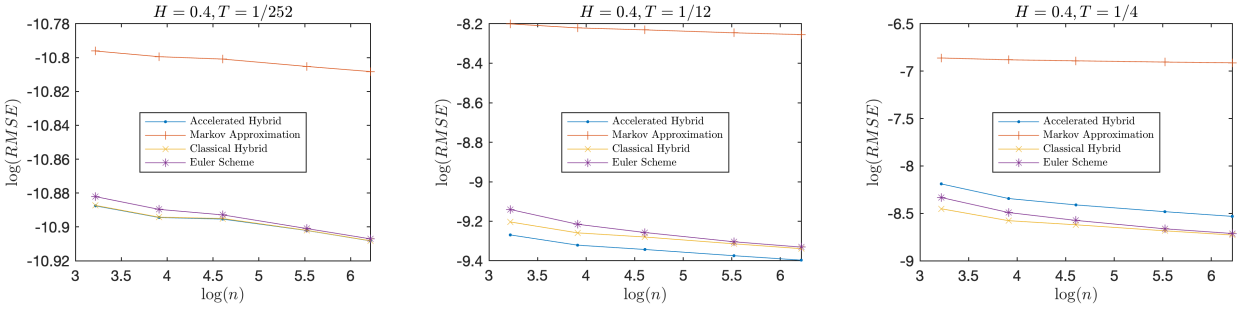
(a) $T = 1/252$ (b) $T = 1/12$ (c) $T = 1/4$

Figure 8: At-the-money European call option pricing errors under $H = 0.2$. RMSE $= \sqrt{\text{standard error}^2 + \text{bias}^2}$. Parameters are chosen as $S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1, \kappa = 2, n_f = 100$.



(a) $T = 1/252$ (b) $T = 1/12$ (c) $T = 1/4$

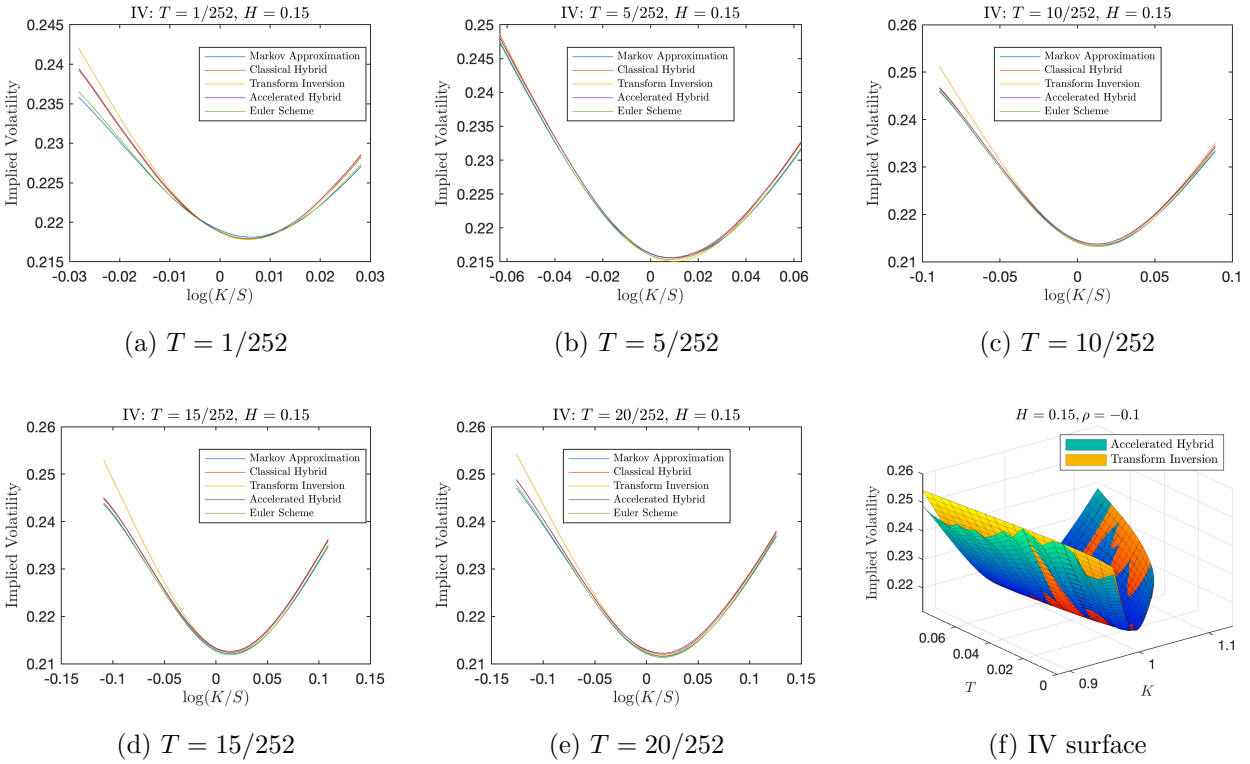
Figure 9: At-the-money European call option pricing errors under $H = 0.3$. RMSE $= \sqrt{\text{standard error}^2 + \text{bias}^2}$. Parameters are chosen as $S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1, \kappa = 2, n_f = 100$.



(a) $T = 1/252$ (b) $T = 1/12$ (c) $T = 1/4$

Figure 10: At-the-money European call option pricing errors under $H = 0.4$. RMSE $= \sqrt{\text{standard error}^2 + \text{bias}^2}$. Parameters are chosen as $S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1, \kappa = 2, n_f = 100$.

same level as other methods, while our scheme's computational effciency significantly outperforms the others (see Table 3).



(a) $T = 1/252$ (b) $T = 5/252$ (c) $T = 10/252$

(d) $T = 15/252$ (e) $T = 20/252$ (f) IV surface

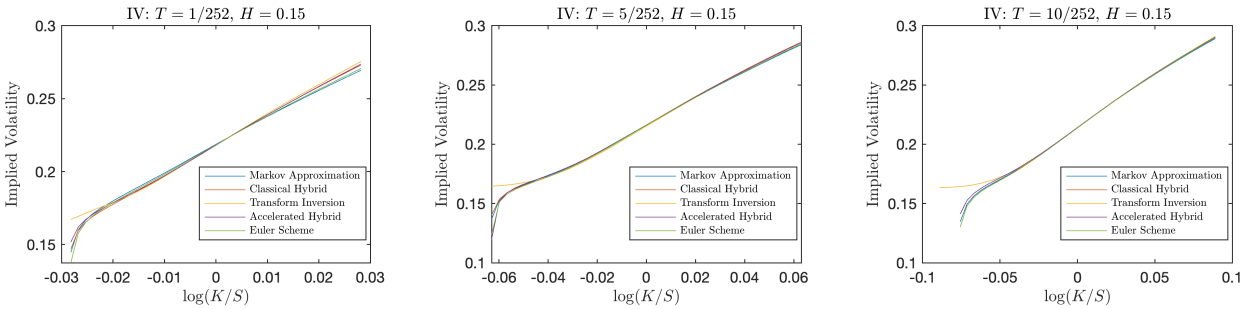
Figure 11: Implied volatility curves for options expiring in 1, 5, 10, and 20 trading days. Parameters are chosen as $H = 0.15, n = 50, n_f = 400, S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.1$, and $\kappa = 2$.

Furthermore, Figure 11 indicates that the implied volatility surface computed via the Accelerated Hybrid Scheme exhibits reduced smoothness in the deep in-the-money region compared to the Fourier transform inversion method. This phenomenon is not unique to our approach but is observed across all simulation-based methods considered in this study. The issue becomes more pronounced when the correlation parameter ρ is set to a highly positive value, such as 0.7, as illustrated in Figure 12, where the implied volatility surface is absent in the deep in-the-money region due to the failure of all simulation methods to produce reliable option price estimates. A plausible explanation is that, for deep in-the-money options, the option price is dominated by its intrinsic value, and the time value component becomes negligible. In Monte Carlo simulations, this results in most simulated paths ending in the money, making the estimation of the small time value component highly sensitive to statistical noise and leading to substantial variance. This challenge can be partially addressed by increasing the number of Monte Carlo paths or employing advanced variance reduction techniques. Notably, this issue does not arise when ρ is strongly negative. For instance, as shown in Figure 13 with

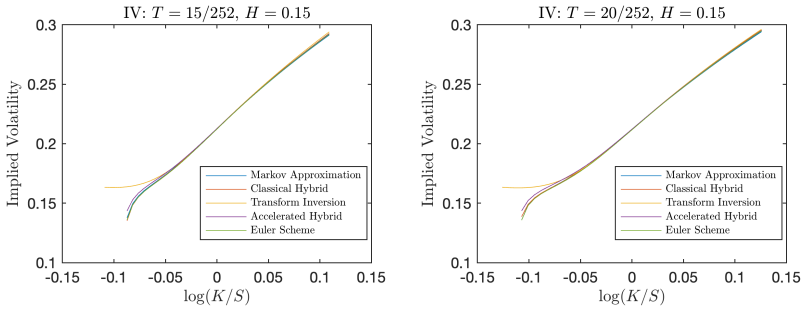
	Markov Approximation	Classical Hybrid	Transform Inversion	Accelerated Hybrid	Euler Scheme
CPU time (s)	104.70	437.07	1879.83	180.70	466.12

Table 3: CPU times for implied volatility calculations under $H = 0.15, \rho = -0.1, n_f = 400, n = 50$. Other parameters are set as $S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0$. The performance results are measured on a MacBook Pro with a 10-core Apple M2 Pro processor and 32 GB RAM.

$\rho = -0.7$, the implied volatility surface remains smooth and accurate even in the deep in-the-money region. A more thorough investigation into this phenomenon is left for future research.



(a) $T = 1/252$ (b) $T = 5/252$ (c) $T = 10/252$

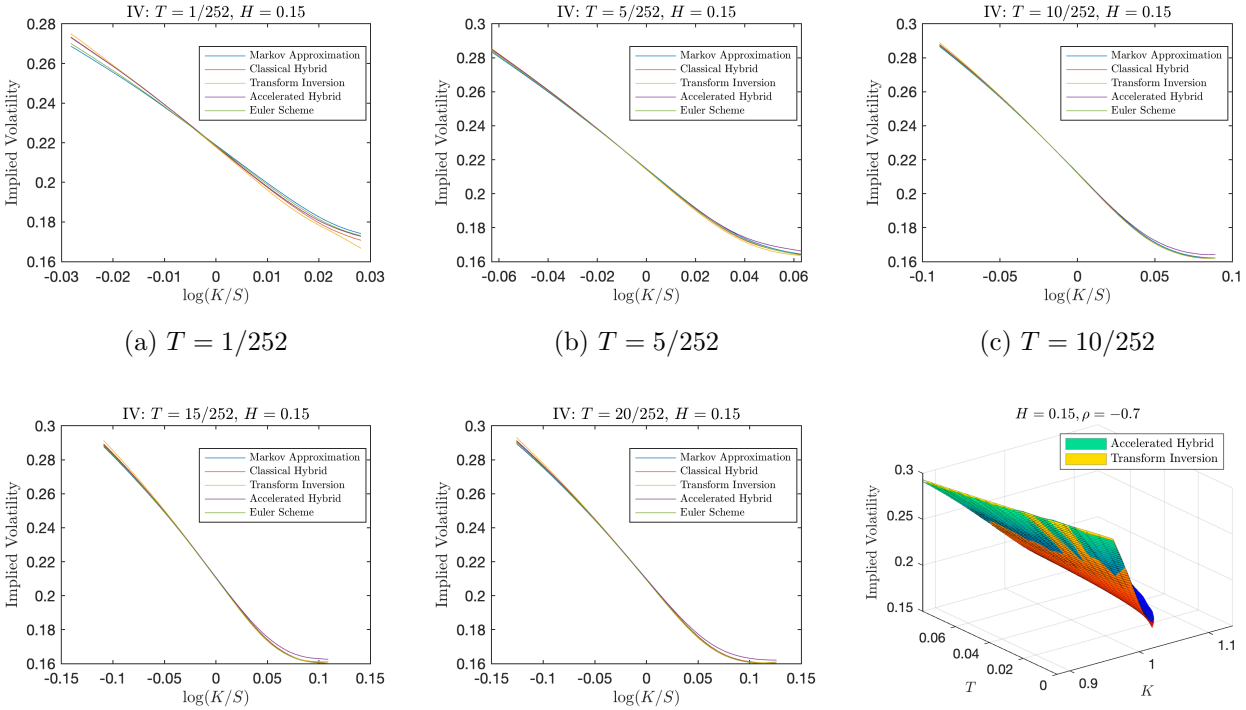


(d) $T = 15/252$ (e) $T = 20/252$

Figure 12: Implied volatility curves for options expiring in 1, 5, 10, and 20 trading days. Parameters are chosen as $H = 0.15, n = 50, n_f = 400, S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = 0.7$, and $\kappa = 2$.

6 Conclusions and discussions

In this paper, we have proposed the Accelerated Hybrid Scheme for simulating rough volatility processes, which combines the advantages of the classical hybrid scheme and the Markovian ap-



(a) $T = 1/252$ (b) $T = 5/252$ (c) $T = 10/252$

(d) $T = 15/252$ (e) $T = 20/252$ (f) IV surface

Figure 13: Implied volatility curves for options expiring in 1, 5, 10, and 20 trading days. Parameters are chosen as $H = 0.15, n = 50, n_f = 400, S_0 = 1, \theta = 0.05, \lambda = 0.1, \nu = 0.3, V_0 = 0.05, r = d = 0, \rho = -0.7$, and $\kappa = 2$.

proximation method. This algorithm is designed to address the computational barrier in simulating stochastic Volterra equations, whose non-Markovian and non-semimartingale nature adds a great deal of redundant computations to classical simulation algorithms. By exploiting a precomputation step based on the Markovian approximation and fast Fourier transform techniques, our approach reduces the computational complexity and improves numerical stability.

The theoretical properties of the Accelerated Hybrid Scheme is well-established. In section 4, we have shown that the algorithm converges strongly to the exact solution of rough volatility processes with globally Lipschitz coefficients b and σ. The strong order of convergence is H with respect to the discretization time steps, making it a rather efficient method for simulating rough volatility processes. In Section 5, the Accelerated Hybrid Scheme is also shown to be robust to non-Lipschitz coefficients. The numerical experiments on the rough Heston model demonstrate that the Accelerated Hybrid Scheme is promising for coefficient $\sigma(x) = \nu\sqrt{x}$, which is non-Lipschitz at $x = 0$.

As a result, the Accelerated Hybrid Scheme offers a significant improvement in simulation speed while preserving accuracy in pricing and implied volatility estimation tasks for rough volatility models. This work opens up new possibilities for the efficient simulation of rough volatility processes, which are essential in industrial applications. Future research directions include the theoretical extension of the Accelerated Hybrid Scheme to more general rough volatility models (e.g., with non-Lipschitz coefficients), the development of a more efficient precomputation step, and the application of the scheme to other financial instruments.

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Appendix A Proof of Theorem 1

Before proving Theorem 1, we introduce the following lemma that establishes the tightness of the Markovian approximation process.

Lemma 1. *Let $\{X_t^{n_f} : 0 \le t \le T\}$ be the Markovian approximation of the process defined by (4) with the kernel approximate $\widehat{K}(t) = \sum_{i=1}^{n_f} w_i e^{-x_i t}$ and $\{(w_i, x_i)\}_{i=1}^{n_f}$ given by (22). Then, for any $t, h \ge 0$ and $t + h \le T$, there exists a constant $C > 0$ such that*

$$\mathbb{E}|X_{t+h}^{n_f} - X_t^{n_f}|^2 \le Ch^{2H}.$$

Proof. By definition of $\{X_t^{n_f} : 0 \le t \le T\}$ and $(a+b)^2 \le 2a^2 + 2b^2$, we have

$$\begin{aligned}\mathbb{E}|X_{t+h}^{n_f} - X_t^{n_f}|^2 &\le 2\mathbb{E}\left|\int_0^{t+h} \widehat{K}(t+h-s)b(X_s^{n_f})\,ds - \int_0^t \widehat{K}(t-s)b(X_s^{n_f})\,ds\right|^2 \\ &\quad + 2\mathbb{E}\left|\int_0^{t+h} \widehat{K}(t+h-s)\sigma(X_s^{n_f})\,dW_s - \int_0^t \widehat{K}(t-s)\sigma(X_s^{n_f})\,dW_s\right|^2 =: 2I_1 + 2I_2.\end{aligned}$$

Using Jensen's inequality, Fubini theorem and linear growth condition, we have

$$\begin{aligned}I_1 &\le 2\mathbb{E}\left|\int_0^t (\widehat{K}(t+h-s) - \widehat{K}(t-s))b(X_s^{n_f})\,ds\right|^2 + 2\mathbb{E}\left|\int_t^{t+h} \widehat{K}(t+h-s)b(X_s^{n_f})\,ds\right|^2 \\ &\le 2t\mathbb{E}\left[\int_0^t |\widehat{K}(t+h-s) - \widehat{K}(t-s)|^2 |b(X_s^{n_f})|^2\,ds\right] + 2h\mathbb{E}\left[\int_t^{t+h} |\widehat{K}(t+h-s)|^2 |b(X_s^{n_f})|^2\,ds\right] \\ &\le 2tC_b\mathbb{E}\left[\int_0^t |\widehat{K}(t+h-s) - \widehat{K}(t-s)|^2 (1+|X_s^{n_f}|)^2\,ds\right] + 2hC_b\mathbb{E}\left[\int_t^{t+h} |\widehat{K}(t+h-s)|^2 (1+|X_s^{n_f}|)^2\,ds\right] \\ &\le 2TC_b\left(\sup_{0\le s\le T}\mathbb{E}[(1+|X_s^{n_f}|)^2]\right)\left(\int_0^t |\widehat{K}(t+h-s) - \widehat{K}(t-s)|^2\,ds + \int_t^{t+h} |\widehat{K}(t+h-s)|^2\,ds\right) \\ &\le 2TC_b\left(\sup_{0\le s\le T}\mathbb{E}[(1+|X_s^{n_f}|)^2]\right)\left(\int_0^{T-h} |\widehat{K}(h+s) - \widehat{K}(s)|^2\,ds + \int_0^h |\widehat{K}(s)|^2\,ds\right) \\ &\le 2TC_1\left(\int_0^{T-h} |\widehat{K}(h+s) - \widehat{K}(s)|^2\,ds + \int_0^h |\widehat{K}(s)|^2\,ds\right),\end{aligned}$$

where $C_1 > 0$ is a constant, since

$$\sup_{0\le t\le T}\mathbb{E}[|X_t^{n_f}|^p] < \infty, \quad p \ge 1,$$

for a given n_f. Likewise, we can obtain a similar bound for I_2 by Itô isometry, which is given by

$$\begin{aligned}I_2 &\le 2\mathbb{E}\left[\int_0^t |\widehat{K}(t+h-s) - \widehat{K}(t-s)|^2 |\sigma(X_s^{n_f})|^2\,ds\right] + 2\mathbb{E}\left[\int_t^{t+h} |\widehat{K}(t+h-s)|^2 |\sigma(X_s^{n_f})|^2\,ds\right] \\ &\le 2C_2\left(\int_0^{T-h} |\widehat{K}(h+s) - \widehat{K}(s)|^2\,ds + \int_0^h |\widehat{K}(s)|^2\,ds\right).\end{aligned}$$

Combining results for I_1 and I_2, we have for any given $T > 0, n_f \geq 1$ that

$$\mathbb{E}|X_{t+h}^{n_f} - X_t^{n_f}|^2 \leq 4(C_1 T + C_2) \left(\int_0^{T-h} |\widehat{K}(h+s) - \widehat{K}(s)|^2 \, ds + \int_0^h |\widehat{K}(s)|^2 \, ds \right),$$

Then, from Section 8.6.1 and Lemma 7.13 in Abi Jaber (2018), we have

$$\sup_{n_f \geq 1} \left(\int_0^{T-h} |\widehat{K}(h+s) - \widehat{K}(s)|^2 \, ds + \int_0^h |\widehat{K}(s)|^2 \, ds \right) \leq C_K h^{2H},$$

which gives that

$$\mathbb{E}|X_{t+h}^{n_f} - X_t^{n_f}|^2 \leq C h^{2H},$$

for some constant $C > 0$.

□

Proof of Theorem 1. Let $T \in \mathbb{R}_+$ be fixed. We can write the approximation $\widetilde{X}_T$ as

$$\begin{aligned}\widetilde{X}_T = X_0 &+ \sum_{i=1}^{n} b\big(\widehat{X}_{t_{n-i}}^{n_f}\big) \int_{(n-i)T/n}^{(n-i+1)T/n} K(T-s) \, ds \\ &+ \sum_{i=1}^{\kappa} \sigma\big(\widehat{X}_{t_{n-i}}^{n_f}\big) \int_{(k-i)T/n}^{(k-i+1)T/n} K(T-s) \, dW_s + \sum_{i=\kappa+1}^{k} K\Big(\frac{b_i T}{n}\Big) \sigma\big(\widehat{X}_{t_{k-i}}^{n_f}\big) \Delta W_{k-i},\end{aligned}$$

and we introduce the auxillary processes

$$\bar{X}_T = X_0 + \int_0^T K(T-s) b(X_s^{n_f}) \, ds + \int_0^T K(T-s) \sigma(X_s^{n_f}) \, dW_s,$$

and

$$\begin{aligned}\bar{X}_T^n = X_0 &+ \sum_{i=1}^{n} b\big(X_{t_{n-i}}^{n_f}\big) \int_{(n-i)T/n}^{(n-i+1)T/n} K(T-s) \, ds \\ &+ \sum_{i=1}^{\kappa} \sigma\big(X_{t_{n-i}}^{n_f}\big) \int_{(k-i)T/n}^{(k-i+1)T/n} K(T-s) \, dW_s + \sum_{i=\kappa+1}^{k} K\Big(\frac{b_i T}{n}\Big) \sigma\big(X_{t_{k-i}}^{n_f}\big) \Delta W_{k-i}.\end{aligned}$$

By $(a+b+c)^2 \leq 3a^2 + 3b^2 + 3c^2$, we have

$$\mathbb{E}|X_T - \widetilde{X}_T|^2 \leq 3\mathbb{E}|X_T - \bar{X}_T|^2 + 3\mathbb{E}|\bar{X}_T - \bar{X}_T^n|^2 + 3\mathbb{E}|\bar{X}_T^n - \widetilde{X}_T|^2.$$

In what follows, we show that each term on the right-hand side converges to 0 as $n_f \to \infty$ and $n \to \infty$.

Let us analyze $\mathbb{E}|X_T - \bar{X}_T|^2$ first:

$$\mathbb{E}|X_T - \bar{X}_T|^2 = \mathbb{E}\left| \int_0^T K(T-s) \left(b(X_s) - b(X_s^{n_f}) \right) ds + \int_0^T K(T-s) \left(\sigma(X_s) - \sigma(X_s^{n_f}) \right) dW_s \right|^2$$

$$\leq 2\mathbb{E}\left|\int_0^T K(T-s)\left(b(X_s)-b(X_s^{n_f})\right)ds\right|^2+2\mathbb{E}\left|\int_0^T K(T-s)\left(\sigma(X_s)-\sigma(X_s^{n_f})\right)dW_s\right|^2$$

By applying Cauchy-Schwarz inequality to the first term, Itô isometry to the second term, and using Fubini theorem, we have

$$\begin{aligned}\mathbb{E}|X_T-\bar{X}_T|^2 &\leq 2T\mathbb{E}\left[\int_0^T K(T-s)^2\left|b(X_s)-b(X_s^{n_f})\right|^2 ds\right]+2\mathbb{E}\left[\int_0^T K(T-s)^2\left|\sigma(X_s)-\sigma(X_s^{n_f})\right|^2 ds\right]\\ &= 2T\int_0^T K(T-s)^2\mathbb{E}\left|b(X_s)-b(X_s^{n_f})\right|^2 ds+2\int_0^T K(T-s)^2\mathbb{E}\left|\sigma(X_s)-\sigma(X_s^{n_f})\right|^2 ds.\end{aligned}$$

Under Assumption 2, there exists a constant $L>0$ such that

$$\begin{aligned}\mathbb{E}\left|b(X_s)-b(X_s^{n_f})\right|^2 &\leq L\cdot\mathbb{E}\left|X_s-X_s^{n_f}\right|^2,\\ \mathbb{E}\left|\sigma(X_s)-\sigma(X_s^{n_f})\right|^2 &\leq L\cdot\mathbb{E}\left|X_s-X_s^{n_f}\right|^2.\end{aligned}$$

Hence, we have

$$\begin{aligned}\mathbb{E}|X_T-\bar{X}_T|^2 &\leq (2T+1)L\int_0^T K(T-s)^2\mathbb{E}\left|X_s-X_s^{n_f}\right|^2 ds\\ &\leq\left(\sup_{0\leq t\leq T}\mathbb{E}\left|X_s-X_s^{n_f}\right|^2\right)\cdot(2T+1)L\int_0^T K(T-s)^2 ds\end{aligned}$$

By Proposition 3, we have

$$\sup_{0\leq t\leq T}\mathbb{E}\left|X_s-X_s^{n_f}\right|^2\leq C\sup_{0\leq t\leq T}\int_0^t|K(s)-\widehat{K}(s)|^2 ds=C\|K-\widehat{K}\|_{2,T}^2,$$

where $C>0$ is a constant, and $\widehat{K}(u)=\sum_{i=1}^{n_f}w_i e^{-x_i u}$ is a kernel approximation following (22). By (23), we have $\|K-\widehat{K}\|_{2,T}\to 0$ as $n_f\to\infty$. Therefore, $\mathbb{E}|X_T-\bar{X}_T|^2\to 0$ as $n_f\to\infty$.

Next, we analyze $\mathbb{E}|\bar{X}_T-\bar{X}_T^n|^2$. Likewise, we introduce an ancillary term

$$\check{X}_T^n=X_0+\sum_{i=1}^n b\big(X_{t_{n-i}}^{n_f}\big)\int_{(n-i)T/n}^{(n-i+1)T/n}K(T-s)\,ds+\sum_{i=1}^n\sigma\big(X_{t_{n-i}}^{n_f}\big)\int_{(k-i)T/n}^{(k-i+1)T/n}K(T-s)\,dW_s,$$

and split the error term as

$$\mathbb{E}|\bar{X}_T-\bar{X}_T^n|^2\leq 2\mathbb{E}|\bar{X}_T-\check{X}_T^n|^2+2\mathbb{E}|\check{X}_T^n-\bar{X}_T^n|^2.$$

Using Jensen's inequality, Itô isometry, Lipschitz continuity of b,σ, and Lemma 1, we have

$$\begin{aligned}\mathbb{E}|\bar{X}_T-\check{X}_T^n|^2 \leq\ & 2\mathbb{E}\left|\sum_{i=1}^n\int_{(n-i)T/n}^{(n-i+1)T/n}K(T-s)(b(X_s^{n_f})-b(X_{t_{n-i}}^{n_f}))\,ds\right|^2\\ &+2\mathbb{E}\left|\sum_{i=1}^n\int_{(n-i)T/n}^{(n-i+1)T/n}K(T-s)(\sigma(X_s^{n_f})-\sigma(X_{t_{n-i}}^{n_f}))\,dW_s\right|^2\end{aligned}$$

$$
\begin{aligned}
&\leq 2T\int_0^T K(T-s)^2\mathbb{E}|b(X_s^{n_f})-b(X_{\eta(s)}^{n_f})|^2\,ds+2\int_0^T K(T-s)^2\mathbb{E}|\sigma(X_s^{n_f})-\sigma(X_{\eta(s)}^{n_f})|^2\,ds\\
&\leq (2T+1)L\cdot\left(\sup_{0\leq s\leq T}\mathbb{E}|X_s^{n_f}-X_{\eta(s)}^{n_f}|^2\right)\int_0^T K(u)^2\,du\\
&\leq (2T+1)LC\left(\frac{T}{n}\right)^{2H}\int_0^T K(u)^2\,du, \qquad \text{(A.1)}
\end{aligned}
$$

where $\eta(s)=\frac{(n-i)T}{n}$ for $s\in[\frac{(n-i)T}{n},\frac{(n-i+1)T}{n}),\ i=1,2,\dots n$, $L>0$ is defined as above, $C>0$ is the constant from Lemma 1. (A.1) indicates that

$$
\mathbb{E}|\bar{X}_T-\check{X}_T^n|^2\to 0,\quad n\to\infty
$$

Likewise, we can derive the upper bound of $\mathbb{E}|\check{X}_T^n-\bar{X}_T^n|^2$ as follows.

$$
\begin{aligned}
\mathbb{E}|\check{X}_T^n-\bar{X}_T^n|^2&=\mathbb{E}\left|\sum_{i=\kappa+1}^n\int_{(n-i)T/n}^{(n-i+1)T/n}\left(K\left(\frac{b_iT}{n}\right)-K(T-s)\right)\sigma(X_{t_{n-i}}^{n_f})\,dW_s\right|^2\\
&=\sum_{i=\kappa+1}^n\int_{(n-i)T/n}^{(n-i+1)T/n}\left(K\left(\frac{b_iT}{n}\right)-K(T-s)\right)^2\mathbb{E}|\sigma(X_{t_{n-i}}^{n_f})|^2\,ds\\
&\leq\frac{\widetilde{C}_\sigma\sup_{0\leq s\leq T}\mathbb{E}(1+|X_s^{n_f}|)^2}{\Gamma(H+\frac{1}{2})^2}\left(\frac{T}{n}\right)^{2H}\sum_{i=\kappa+1}^n\int_{i-1}^i(b_i^{H-\frac{1}{2}}-y^{H-\frac{1}{2}})^2\,dy\\
&\leq C_{\sigma,H}\left(\frac{T}{n}\right)^{2H}\sum_{i=\kappa+1}^n\int_{i-1}^i(b_i^{H-\frac{1}{2}}-y^{H-\frac{1}{2}})^2\,dy \qquad \text{(A.2)}
\end{aligned}
$$

where $C_{\sigma,H}>0$ is a constant related to σ,H and the upper bound of $\sup_{0\leq s\leq T}\mathbb{E}[|X_s^{n_f}|^p]$ for $p>0$ (which is finite by Section 7.5.1, Abi Jaber (2018)). Minimizing $\int_{i-1}^i(b_i^{H-\frac{1}{2}}-y^{H-\frac{1}{2}})^2dy$ above with respect to b_i, we have

$$
b_i=\left(\frac{i^{H+\frac{1}{2}}-(i-1)^{H+\frac{1}{2}}}{H+\frac{1}{2}}\right)^{\frac{1}{H-\frac{1}{2}}}. \qquad \text{(A.3)}
$$

By Theorem 2.5, Bennedsen et al. (2017), we have

$$
\sum_{i=\kappa+1}^\infty\int_{i-1}^i(b_i^{H-\frac{1}{2}}-y^{H-\frac{1}{2}})^2\,dy<\infty,
$$

which directly leads to

$$
\mathbb{E}|\check{X}_T^n-\bar{X}_T^n|^2\to 0,\quad n\to\infty.
$$

Finally, we analyze $\mathbb{E}|\bar{X}_T^n-\widetilde{X}_T|^2$, which is given by

$$
\mathbb{E}|\bar{X}_T^n-\widetilde{X}_T|^2=\mathbb{E}\left|\sum_{i=1}^n(b(X_{t_{n-i}}^{n_f})-b(\widehat{X}_{t_{n-i}}^{n_f}))\int_{(n-i)T/n}^{(n-i+1)T/n}K(T-s)\,ds\right|^2
$$

$$
\begin{aligned}
&+ \mathbb{E}\left|\sum_{i=1}^{\kappa}(\sigma(X^{n_f}_{t_{n-i}}) - \sigma(\widehat{X}^{n_f}_{t_{n-i}}))\int_{(n-i)T/n}^{(n-i+1)T/n} K(T-s)\,dW_s\right|^2 \\
&+ \mathbb{E}\left|\sum_{i=\kappa+1}^{n} K\left(\frac{b_i T}{n}\right)(\sigma(X^{n_f}_{t_{n-i}}) - \sigma(X^{n_f}_{t_{n-i}}))\int_{(n-i)T/n}^{(n-i+1)T/n} dW_s\right|^2 \\
&=: J_1 + J_2 + J_3. \qquad \text{(A.4)}
\end{aligned}
$$

Likewise, by Jensen's inequality and Lipschitz continuity,

$$
\begin{aligned}
J_1 &= \mathbb{E}\left|\int_0^T K(T-s)(b(X^{n_f}_{\eta(s)}) - b(\widehat{X}^{n_f}_{\eta(s)}))\,ds\right|^2 \\
&\le T\int_0^T K(T-s)^2\mathbb{E}|b(X^{n_f}_{\eta(s)}) - b(\widehat{X}^{n_f}_{\eta(s)})|^2\,ds \\
&\le TL\cdot\left(\max_{i\in\{0,\dots,n-1\}}\mathbb{E}|X^{n_f}_{t_i} - \widehat{X}^{n_f}_{t_i}|^2\right)\int_0^T K(u)^2\,du \qquad \text{(A.5)}
\end{aligned}
$$

where $L > 0$ is the Lipschitz constant, $\eta(\cdot)$ is defined as above. Similarly, by Itô isometry and Lipschitz continuity, we have

$$
\begin{aligned}
J_2 + J_3 &= \sum_{i=1}^{\kappa}\int_{(n-i)T/n}^{(n-i+1)T/n} K(T-s)^2\mathbb{E}|\sigma(X^{n_f}_{t_{n-i}}) - \sigma(\widehat{X}^{n_f}_{t_{n-i}})|^2\,ds \\
&\quad + \sum_{i=\kappa+1}^{n} K\left(\frac{b_i T}{n}\right)^2\mathbb{E}|\sigma(X^{n_f}_{t_{n-i}}) - \sigma(\widehat{X}^{n_f}_{t_{n-i}})|^2\left(\frac{T}{n}\right) \\
&\le L\cdot\left(\max_{i\in\{0,\dots,n-1\}}\mathbb{E}|X^{n_f}_{t_i} - \widehat{X}^{n_f}_{t_i}|^2\right)\left(\sum_{i=1}^{\kappa}\int_{(i-1)T/n}^{iT/n} K(u)^2du + \frac{T}{n}\sum_{i=\kappa+1}^{n} K\left(\frac{b_i T}{n}\right)^2\right) \qquad \text{(A.6)}
\end{aligned}
$$

It is easy to show that $\frac{T}{n}\sum_{i=\kappa+1}^{n} K\left(\frac{b_i T}{n}\right)^2 < \infty$ as $n\to\infty$ for any sequence $b_i \in [i-1, i],\ i = 1,\dots n$. As the discretization scheme (26)-(27) is inherently the same as the Euler scheme (8) (Theorem 4.2, Alfonsi and Kebaier (2023)) with kernel $\widehat{K}(u) = \sum_{i=1}^{n_f} w_i e^{-x_i u}$, by Proposition 1,

$$
\max_{i\in\{0,\dots,n-1\}}\mathbb{E}|X^{n_f}_{t_i} - \widehat{X}^{n_f}_{t_i}|^2 \to 0, \quad n\to\infty,
$$

which leads to

$$
\mathbb{E}|\bar{X}^n_T - \widetilde{X}_T|^2 = J_1 + J_2 + J_3 \to 0, \quad n\to\infty.
$$

Combining the convergence results of $\mathbb{E}|X_T - \bar{X}_T|^2$, $\mathbb{E}|\bar{X}_T - \bar{X}^n_T|^2$, and $\mathbb{E}|\bar{X}^n_T - \widetilde{X}_T|^2$, we can therefore conclude that

$$
\mathbb{E}|X_T - \widetilde{X}_T|^2 \to 0, \quad n_f\to\infty, \quad n\to\infty.
$$

□